

Welcome to

Networking and TCP/IP for Beginners Workshop I: Basics

PROSA November 2026

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Slides are available as PDF, https://codeberg.org/kramse

basic-tcpip-beginners-I.tex in the repo security-courses

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Contact information



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- Run a small network for fun AS57860

You are welcome to drop me an email

Goals for today



- Introduce basic networking and TCP/IP terminology
- Show various network configurations with common protocols
- Describe basics of TCP/IP
- Let you get some hands on with IP protocols
- Connecting to networks bring security risks, sorry – so we will introduce a little security at the end

Photo is NWWC camp at BornHack 2024, next is BornHack July 21. to 28. 2027 https://bornhack.dk

Time schedule



17:00 - 18:15 – Session 0: Introduction and basic network terms, small exercises

18:15 - 18:45 – Break

18:45 - 19:30 – Session 1: Diving deeper into networking and TCP/IP

19:30 - 19:45 – Break

19:45 - 20:30 – Session 2: Try out a more exercises, play with protocols, see devices

20:30 - 21:00 – Activating firewalls, Discussion and questions

During the presentation, in breaks etc. please do connect to the network, play with TCP/IP, switches and routers.

Session 0: Introduction and basic network terms, small exercises



Exercises are completely optional



We will use a combination of your systems, my networking hardware and my systems.
There might be live sniffing done on traffic!

We try to mimic what you would do in your own networks during the exercises.

- Try ping and traceroute
- See your own IP settings
- Borrow a USB Ethernet and connect to a switch or router
- Borrow a router home – only brought a few!

Linux is a toolbox



I will use Linux and participants are welcome to use any operating system of their choice.

Note: even though I talk a lot about Unix, you can definitely run a lot of tools on Windows and Mac OS X.

- Ping and traceroute - they are on every OS
- USB Ethernet work on most operating system
- USB Wi-Fi cards are fun, but some takes a little work to get going
- Basic tools like the Nmap suite is available for Microsoft Windows, Mac OS X, Linux and many others

Command line tools are sometimes used in the slides, as they only show text where a GUI screenshot can be cluttered with a lot of information, feel free to find GUI tools and web sites with same functionality



If you are interested in TCP/IP you are welcome

It is recommended to use virtual machines for the exercises

Network security and most internet related security work has the following requirements:

- Network experience
- TCP/IP principles - often in more detail than a common user
- Programming is an advantage, for automating things
- Some Linux and Unix knowledge is in my opinion a **necessary skill** for infosec work
  – too many new tools to ignore, and lots found at sites like Github and Open Source written for Linux

Course Materials

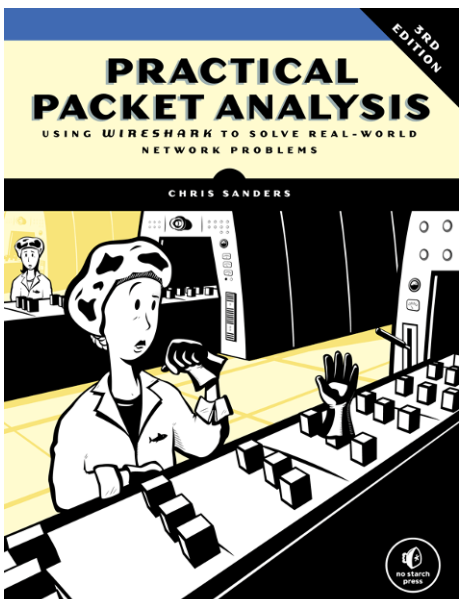


- This material is in multiple parts:
- Slide show - presentation - this file
- Exercises - PDF which is used for this and other workshops
- Additional resources from the internet are linked throughout
- Wikipedia has a LOT of nice pages about IP protocols, for example:

> Transport Layer Security (TLS) is a cryptographic protocol designed to provide communications security over a computer network. The protocol is widely used in applications such as email, instant messaging, and voice over IP, but its use in securing HTTPS remains the most publicly visible.

Source: https://en.wikipedia.org/wiki/Transport_Layer_Security

Book: Practical Packet Analysis (PPA)



Practical Packet Analysis, Using Wireshark to Solve Real-World Network Problems by Chris Sanders, 3rd Edition April 2017, 368 pp. ISBN-13: 978-1-59327-802-1 https://nostarch.com/packetanalysis3

I recommend this book for people new to networking, it has been in HumbleBundle book bundles multiple times



Computer network

In computer science, computer engineering, and telecommunications, **a network** is a **group** of communicating computers and peripherals called hosts, which communicate data to other hosts via communication protocols, as facilitated by networking hardware.

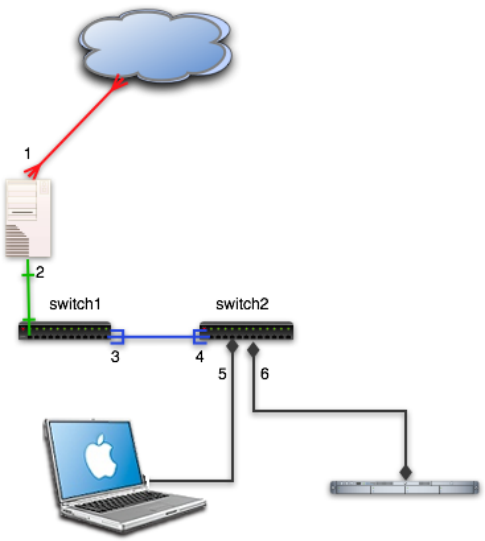
Within a computer network, **hosts** are identified by **network addresses**, which allow networking hardware to locate and identify hosts. Hosts may also have **hostnames**, memorable labels for the host nodes, which can be mapped to a network address using a hosts file or a name server such as **Domain Name Service**. The physical medium that supports information exchange includes **wired media like copper cables, optical fibers, and wireless radio-frequency media**. The arrangement of hosts and hardware within a network architecture is called the **network topology**.[1][2]

Source: https://en.wikipedia.org/wiki/Computer_network

Course Network



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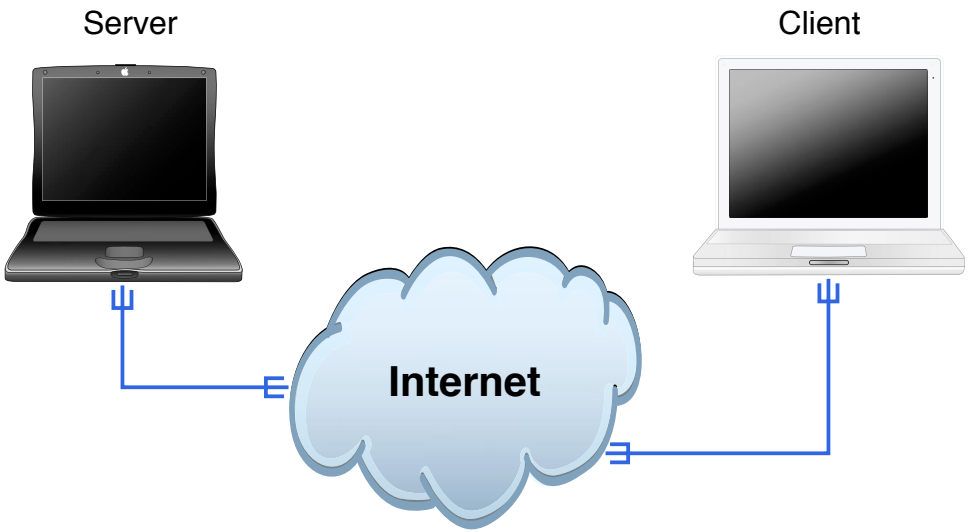
I have a course network with me which has the following information:

- Routers – computer with OpenBSD and portable routers
- Switches and wireless access-points – Unifi AP
- Local LAN wired and wireless – uses something called RFC1918 private addresses

There is also a public network used in examples:

- IPv4 addresses: 185.129.63.0/24 total 256, IPv6 addresses route6: 2a06:d380::/32
- Autonomous system number: AS57860 Zencurity Aps
  https://en.wikipedia.org/wiki/Autonomous_system_(Internet)

Internet Today



Clients and servers, roots in the academic world

Protocols are old, very old! RFC 791: STD 5: Internet Protocol is from September 1981!

Very few protocols were encrypted, today a lot has switched to HTTPS and TLS

Internet is Open Standards!



We reject kings, presidents, and voting.
We believe in rough consensus and running code.
– The IETF credo Dave Clark, 1992.

Request for comments (RFC) – a series of documents spanning decades

RFC, BCP, FYI, informational – first ones from 1969!

Are not updated but status is changed to Obsoleted when new versions are published

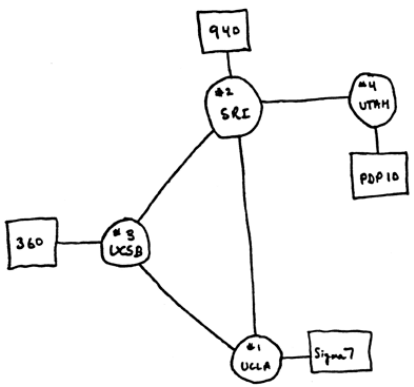
Standards track:
Proposed Standard $\rightarrow$ Draft Standard $\rightarrow$ Standard

Internetworking: history



1961 L. Kleinrock, MIT packet-switching theory
1962 J. C. R. Licklider, MIT - notes
1964 Paul Baran: On Distributed Communications
1969 ARPANET 4 nodes
1971 14 nodes
1973 Design of Internet Protocols started
1973 Email is about 75% of all ARPANET traffic
1974 TCP/IP: Cerf/Kahn: A protocol for Packet Network Interconnection
1983 EUUG $\rightarrow$ DKUUG/DIKU Denmark
1988 About 60.000 systems on the internet - The Morris Worm hits about 10%
2002 About 130 million Internet hosts
2010 IANA reserved blocks 7% (Maj 2010) - http://www.potaroo.net/tools/ipv4/

Internet historically set - anno 1969

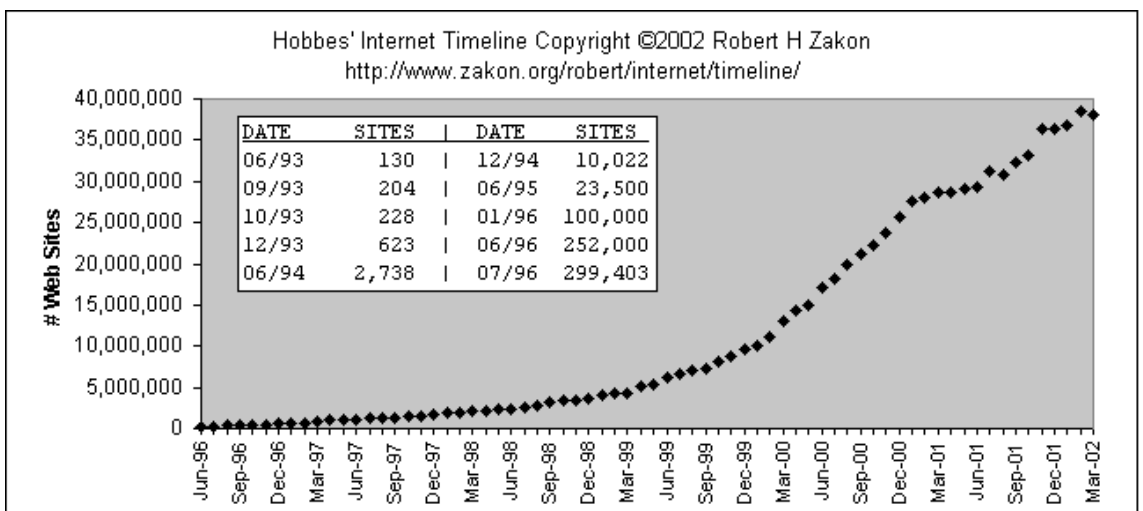


- Node 1: University of California Los Angeles
- Node 2: Stanford Research Institute
- Node 3: University of California Santa Barbara
- Node 4: University of Utah

What are Internet hosts



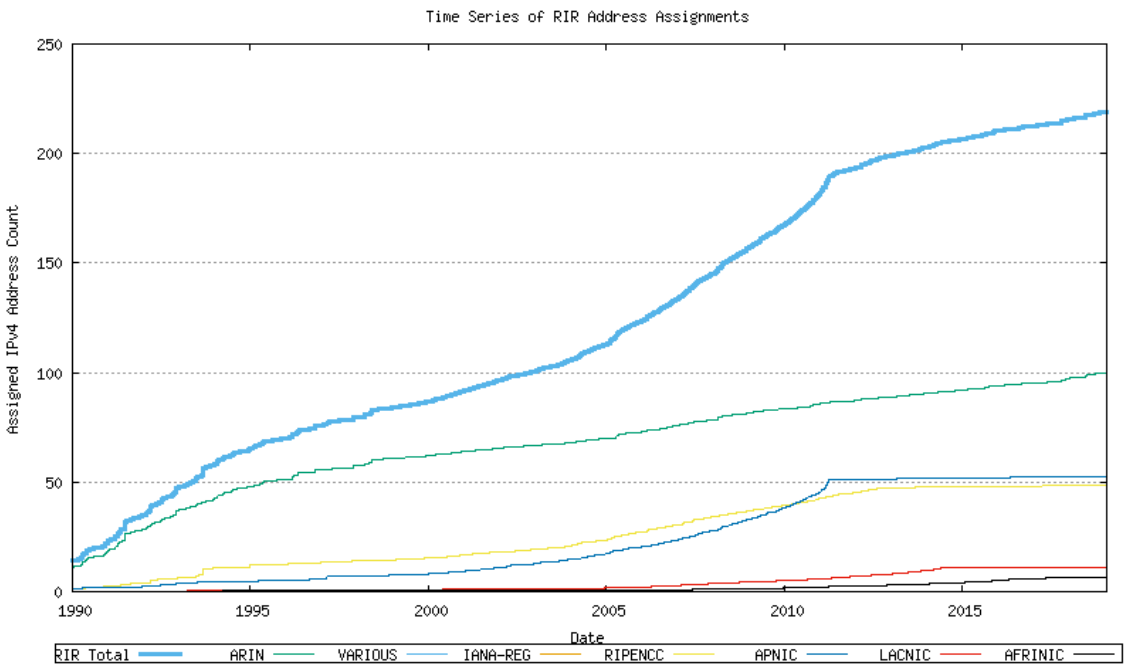
World Wide Web servers



DATE	SITES	DATE	SITES
06/93	130	12/94	10,022
09/93	204	06/95	23,500
10/93	228	01/96	100,000
12/93	623	06/96	252,000
06/94	2,738	07/96	299,403

Source: Hobbes' Internet Timeline v5.6 http://www.zakon.org/robert/internet/timeline/

What are Internet hosts



Cumulative RIR address assignments, per RIR

Source: IPv4 Address Report http://www.potaroo.net/tools/ipv4/

What is the Internet



Communication between humans - currently!

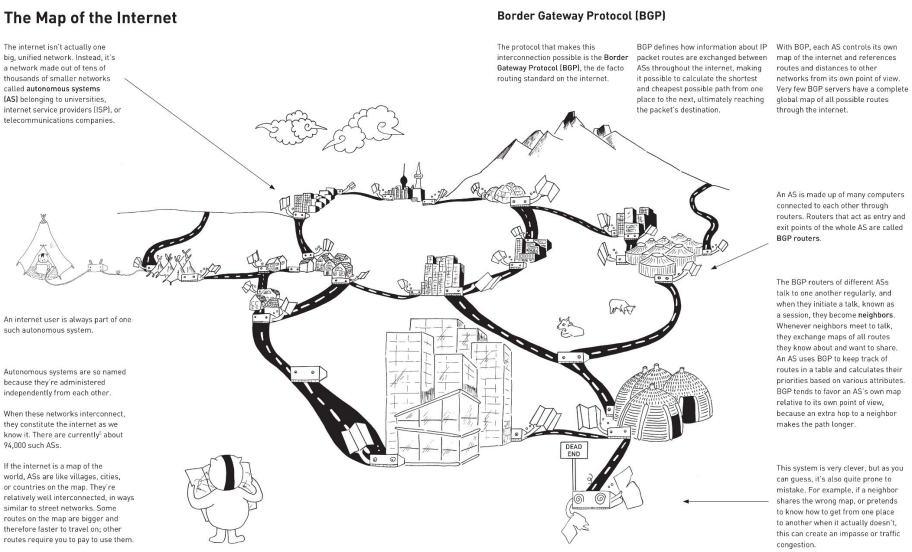
Based on TCP/IP

- best effort – send it off, hope for the best, often it is received at destination
- packet switching (IPv6 calls it packets, not datagram)
- *connection-oriented* TCP
- *connection-less* UDP

RFC-1958:

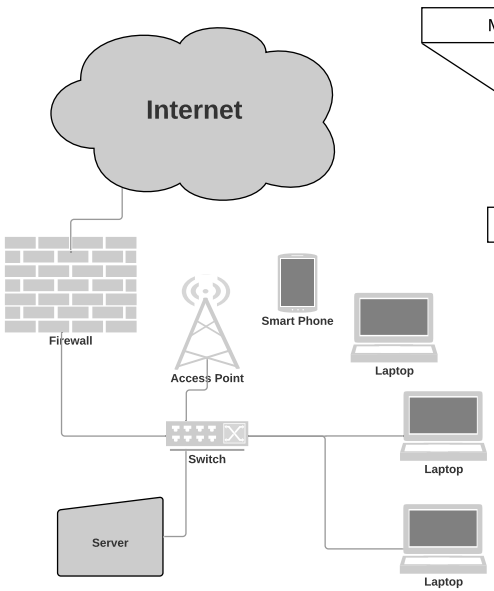
> A good analogy for the development of the Internet is that of constantly renewing the individual streets and buildings of a city, rather than razing the city and rebuilding it. The architectural principles therefore aim to provide a framework for creating cooperation and standards, as a small "spanning set"of rules that generates a large, varied and evolving space of technology.

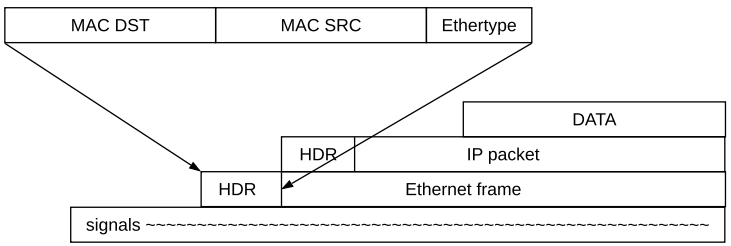
What the internet is actually made up of



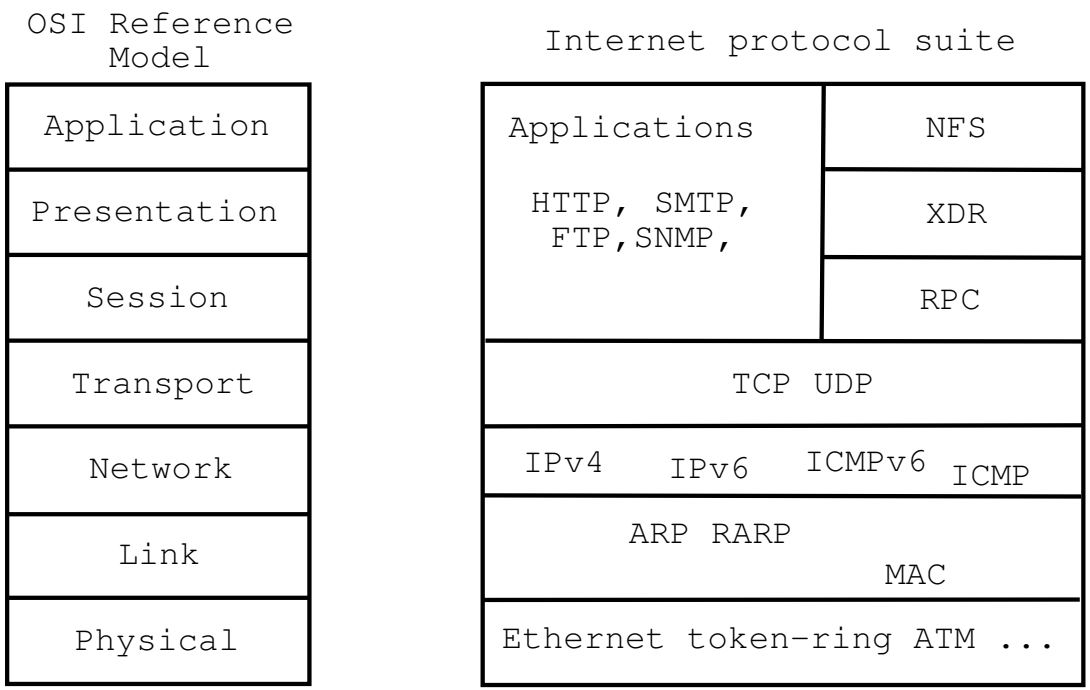
Source: *How the Internet Really Works An Illustrated Guide to protocols, privacy, censorship, and governance* by catnip and article 19 https://catnip.article19.org/ ❤ fun and educational book

Network data at different layers




MAC DST
MAC SRC
Ethertype
DATA
HDR
IP packet
HDR
Ethernet frame
signals ~~~

Protocols: OSI and Internet models



Protocols are rules, like how to say; Hello! Don't worry too much yet, this is an overview!

Recommended network technologies to learn



Small set of recommended networking technologies to learn

Networking: Basic Protocols from the Internet Protocols suite – TCP/IP

- Network Layer: Ethernet, Address Resolution Protocol (ARP), IPv4 and ICMPv4
- Transport Layer: Transmission Control Protocol (TCP) and User Datagram Protocol (UDP)
- Common upper layer: Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), Hypertext Transfer Protocol (HTTP)

- Later add the encrypted/secure versions like Hypertext Transfer Protocol Secure (HTTPS) which uses Transport Layer Security (TLS)
- Later add Wi-Fi and IPv6/ICMPv6 with Neighbor Discovery Protocol (NDP)

Enough to be able to go to other resources, and a connected path from ARP to TLS

How to use the Internet Protocols (IP) with DNS



Names are used by humans when browsing, sending email etc.



The Domain Name System (DNS)



Names are used by humans

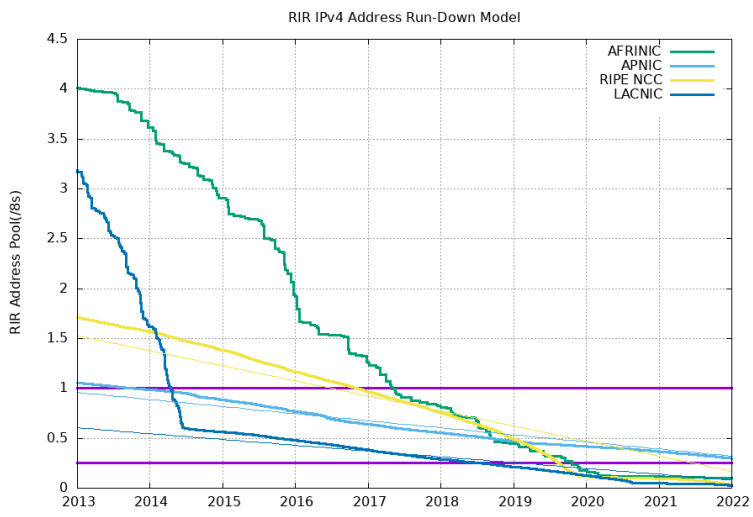
www.kramse.org hlk@kramse.org

Computers use the addresses

```
$ host www.kramse.dk
www.kramse.dk has address 185.129.63.130
www.kramse.dk has IPv6 address 2a06:d380:0:102::80
...
mail.kramse.org has address 185.129.63.250
mail.kramse.org has IPv6 address 2a06:d380:0:102::25
```

This is the main feature of Domain Name System (DNS) – forward and reverse lookups
See more at https://en.wikipedia.org/wiki/Domain_Name_System

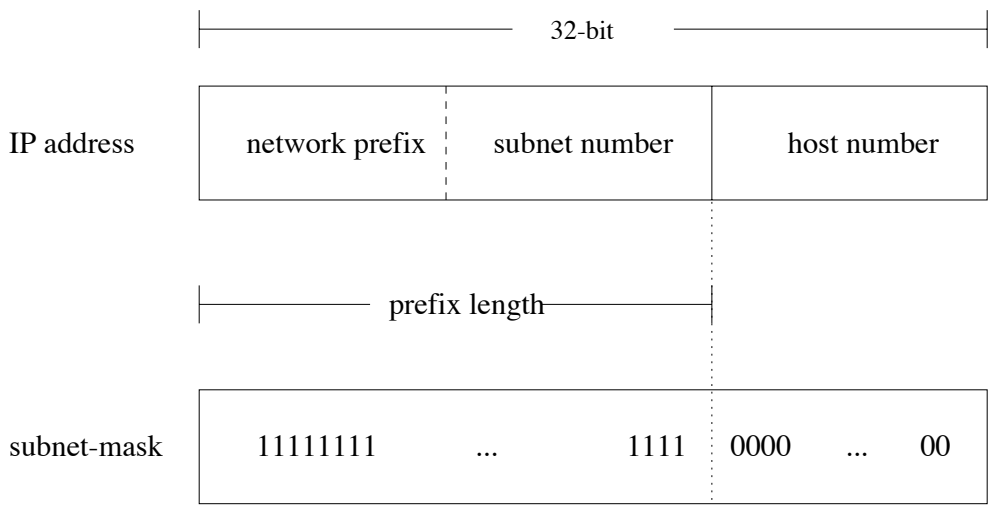
IPv4 ran out of addresses



Source: http://www.potaroo.net/tools/ipv4/

We have essentially run out of IPv4 public addresses, so almost everyone uses Network Address Translation (NAT)

Common Address Space



- Internet is defined by the address space
- IPv4 based on 32-bit addresses, example **dotted decimal** format 10.0.0.1
- IPv6 very similar to IPv4 without NAT, 128-bit addresses in hex ::1, 2a06:d380:0:101::80

Internet ABC was replaced by CIDR



Classful routing

4 class C networks	Inherent subnet mask
192.0.8.0	255.255.255.0
192.0.9.0	255.255.255.0
192.0.10.0	255.255.255.0
192.0.11.0	255.255.255.0

Classless routing CIDR

Supernet	Subnet mask
192.0.8.0	255.255.252.0
	252d=11111100b

Base network/prefix
192.10.8.0/22

- Previously we used classes: A, B, C, D og E See http://en.wikipedia.org/wiki/Classful_network
- This proved to be a bit inflexible and today we use CIDR
- Classless Inter-Domain Routing (CIDR) always use subnet mask which is a number of 1-bit followed by zeroes
- 10.0.0.0/24 means the network 10.0.0.0 with 24 subnet bits (mask 255.255.255.0)
- 2a06:d380:0:101::80/64 means the network with 64-bit prefix length
- **Supernet** is when more networks are **aggregated** into one

RFC-1918 Private Networks



Address Allocation for Private Internets RFC-1918

- 10.0.0.0 - 10.255.255.255 (10/8 prefix)
- 172.16.0.0 - 172.31.255.255 (172.16/12 prefix)
- 192.168.0.0 - 192.168.255.255 (192.168/16 prefix)

Can be used by anyone in their own local area network (LAN)

A typical *home network* would use 192.168.0.0/24 with 192.168.0.1 being the router

Documentation Prefix, IPv6 updates etc.



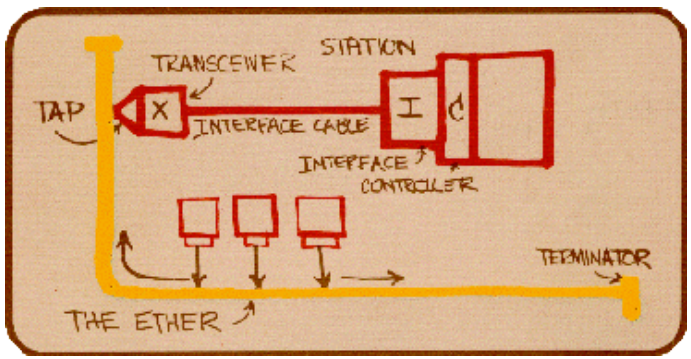
Even documentation has its own prefix, RFC5737:

```
The blocks 192.0.2.0/24 (TEST-NET-1), 198.51.100.0/24 (TEST-NET-2),
and 203.0.113.0/24 (TEST-NET-3) are provided for use in
documentation.
```

IPv6 listed in RFC3849 `2001:DB8::/32` with RFC 9637 (2024) added a larger documentation block, `3fff::/20`

See RFC3330 *Special-Use IPv4 Addresses* which is updated by RFC6890 *Special-Purpose IP Address Registries* which in turn is updated by RFC8190
Use the web version of RFCs to surf back and forth https://www.rfc-editor.org/rfc/rfc8190

Ethernet and Bridges



Original by Robert M. Metcalfe

- Ethernet is a broadcast technology data transmitted into the ether – a cable
- This limits how many devices to connect, but using bridges we can connect segments – which copy between them if needed
- It learns the devices on each side (MAC address)
- See also http://en.wikipedia.org/wiki/ALOHAnet

Hardware addressing



Many LAN hardware devices use *hardware address* – MAC **a8:5e:45:bc:6b:f7**

```
A8-5E-45    (hex)  ASUSTek COMPUTER INC. 27 July 2019
A85E45      (base 16)  ASUSTek COMPUTER INC.
    15,Li-Te Rd., Peitou, Taipei 112, Taiwan
    Taipei  Taiwan  112
    TW
```

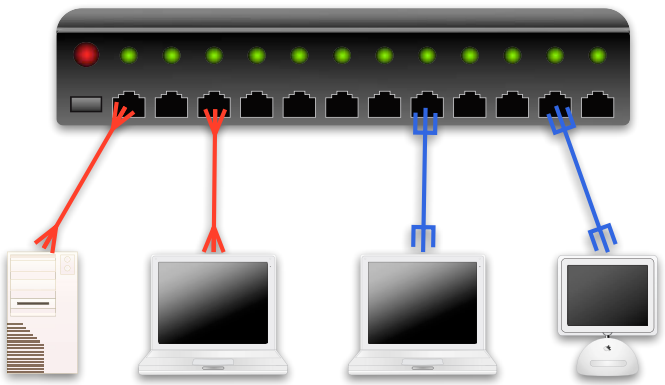
- Network technologies use a layer 2 hardware address
- Typically using 48-bit Media Access Control (MAC) addresses known from Ethernet MAC-48/EUI-48
- First half is assigned to companies – Organizationally Unique Identifier (OUI)
- Using the OUI you can see which producer and roughly when a network chip was produced
- http://standards.ieee.org/regauth/oui/index.shtml

Ethernet, cables



Show link, and activity – blinkenlights

A switch – many bridges combined into one device



- Today we use switches, Don't buy a hub, not even for experimenting or sniffing
- A switch can receive and send data on multiple ports at the same time in the **same subnet**
- Performance only limited by the backplane and switching chips
- Switches can often **route** with the same speed and mirror packets
- **Routing connects subnets** – moving packets between subnets

Wireless



A typical home router would have built-in 802.11 Access-Point (AP) and some Ethernet LAN ports

Wifi network cards



Varenummer: 2225730

TP-Link TL-WN722N

Hi-Speed USB - 802.11b, 802.11g, 802.11n

På lager, 1-2 dages levering
(Billigste fragt: 0 kr.)
Ret land

Køb

120,00 kr.
(96,00 kr.)

4 stk på lager i Århus
0 stk på lager i Viborg
0 stk på lager i København

- Your devices typically have (multiple) wireless chips supporting wireless communication
- Card shown is TP-link TL-WN722N hardware version 2.0 cheap but only support 2.4GHz
- Alfa AWUS036ACH 2.4GHz + 5GHz Dual-Band and high performing
- Both work great in Kali Linux for our purposes, **but are older models by now**

Unfortunately the vendors change models often enough that the above are hard to find. I recommend using your favourite search engine and research which cards work with Kali Linux and airodump-ng.



Wifi standards IEEE 802.11

- 802.11b 11Mbps
- 802.11g 54Mbps
- 802.11n faster
- 802.11i Security enhancements Robust Security Network RSN

New names:
Wi-Fi 6 to identify devices that support 802.11ax technology
Wi-Fi 5 to identify devices that support 802.11ac technology
Wi-Fi 4 to identify devices that support 802.11n technology

Source: http://grouper.ieee.org/groups/802/11/index.html

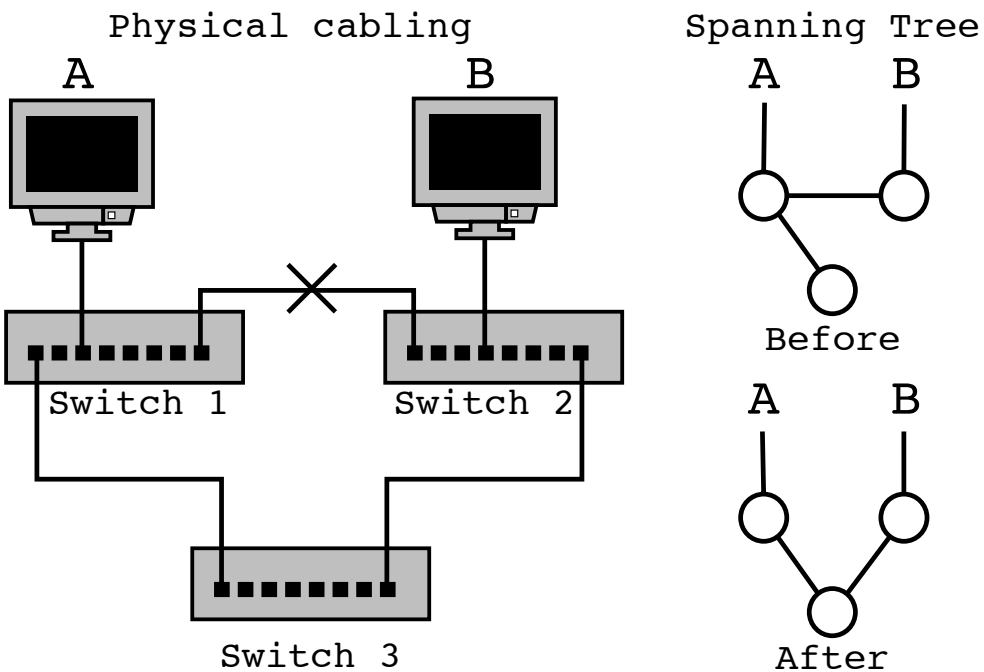
A modern router



- Opal (GL-SFT1200) is a pocket-sized travel router supporting 1200Mbps wireless, Max. 300 Mbps (2.4GHz) + 867 Mbps (5GHz) Fast Wi-Fi Speeds, Powerful CPU with 3 x Gigabit Ports, Excellent Security with VPN – OpenVPN & WireGuard, IPv6

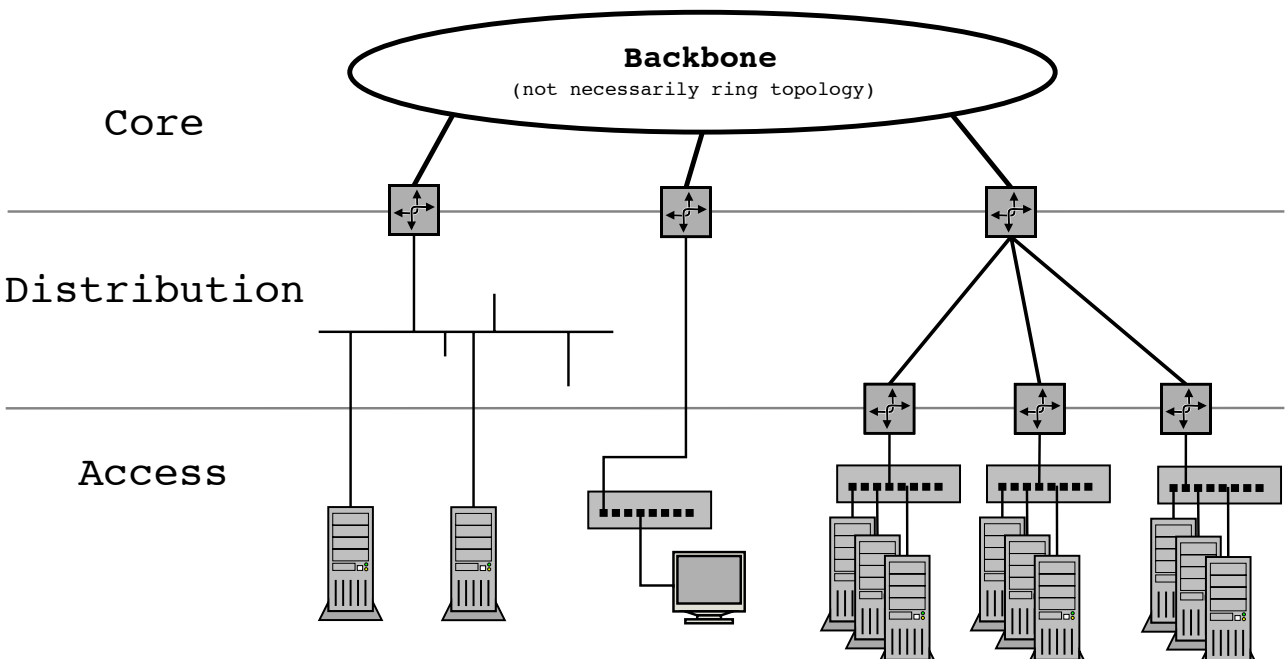
  Source: https://www.gl-inet.com/products/gl-sft1200/

Topologier and Spanning Tree Protocol (STP)



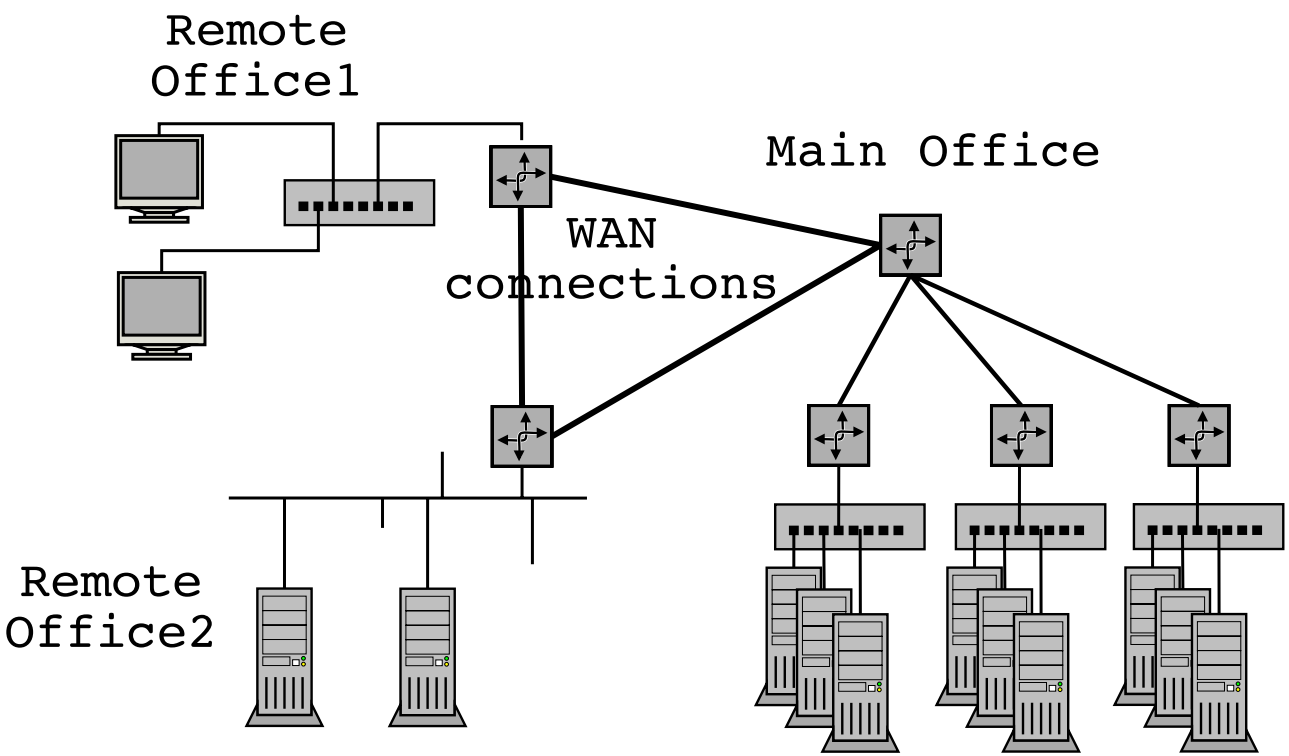
See more in Radia Perlman, *Interconnections: Bridges, Routers, Switches, and Internetworking Protocols*

Core, Distribution and Access net

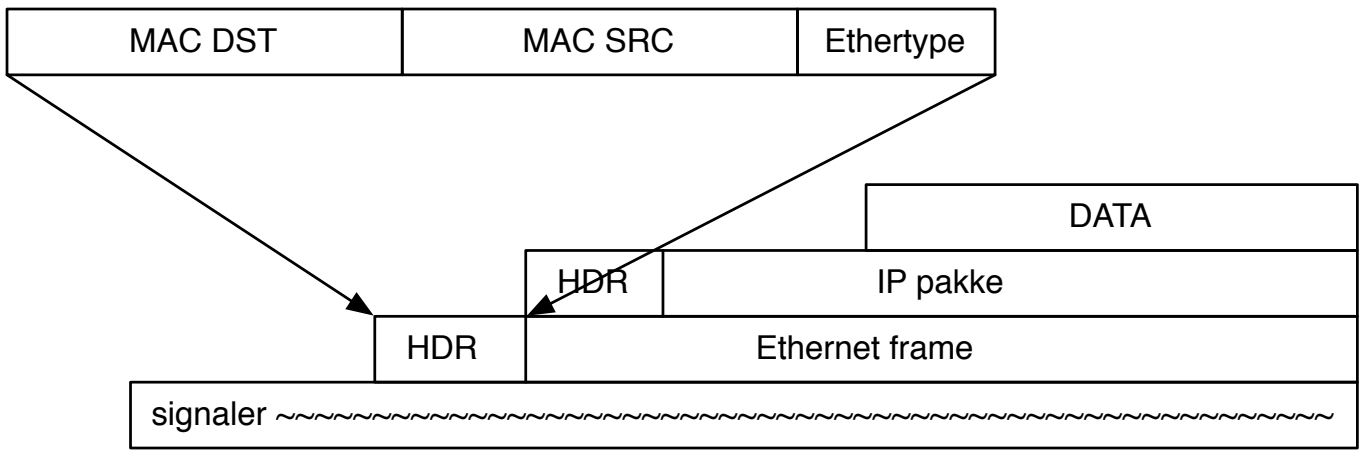


Not always using these terms in modern networking

Bridges and routers across a Wide Area Network (WAN)



Packets of data – postcards of the Internet



Looking deeper data is laid out according to a structure – **frames and packets**

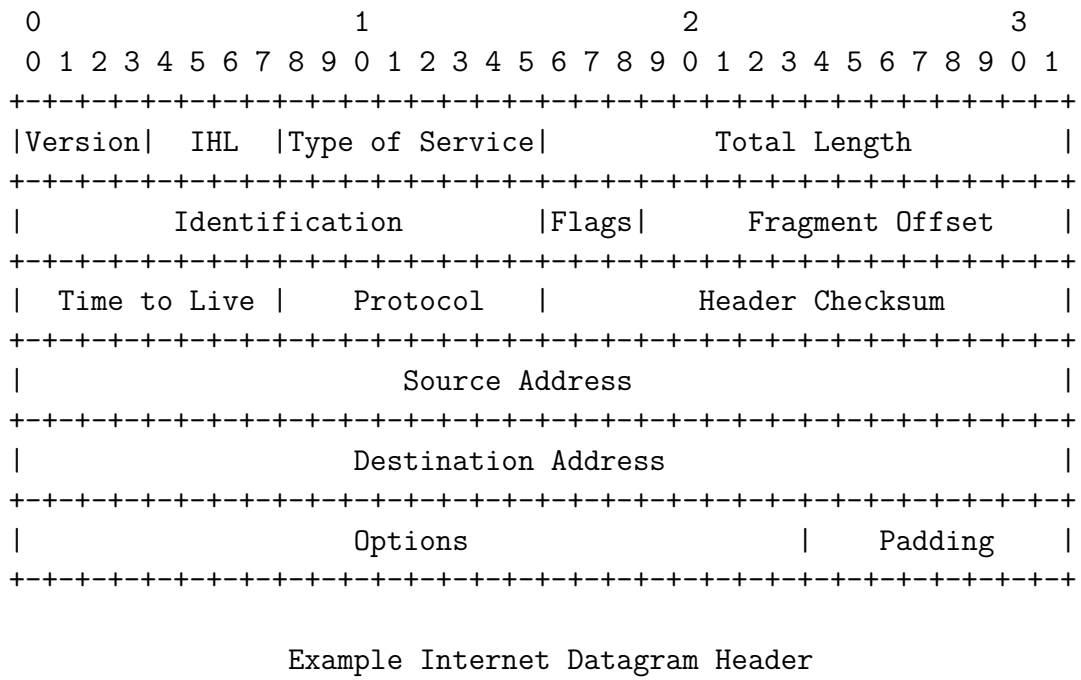
Often a start and end signal of a frame – like Ethernet

Today we talk about packets of 1500 bytes which is common in Ethernet

Operating system masks a lot of complexity

IPv4 header - RFC-791 september 1981



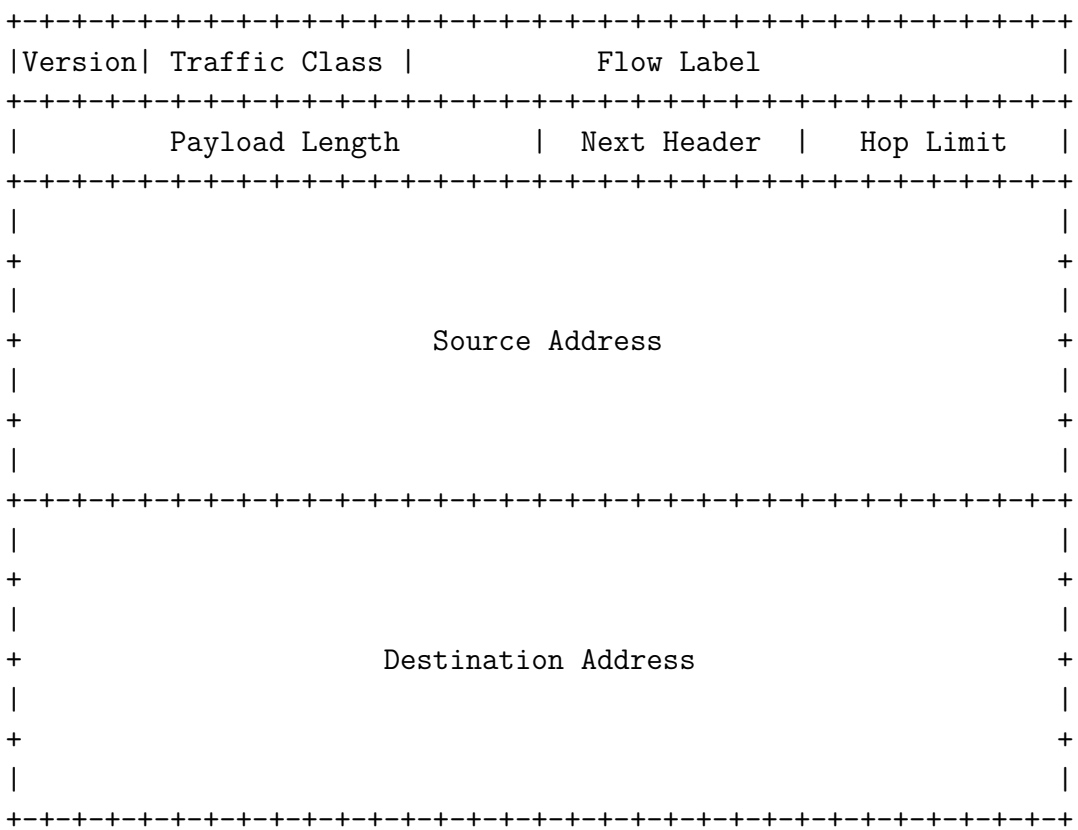


Example Internet Datagram Header

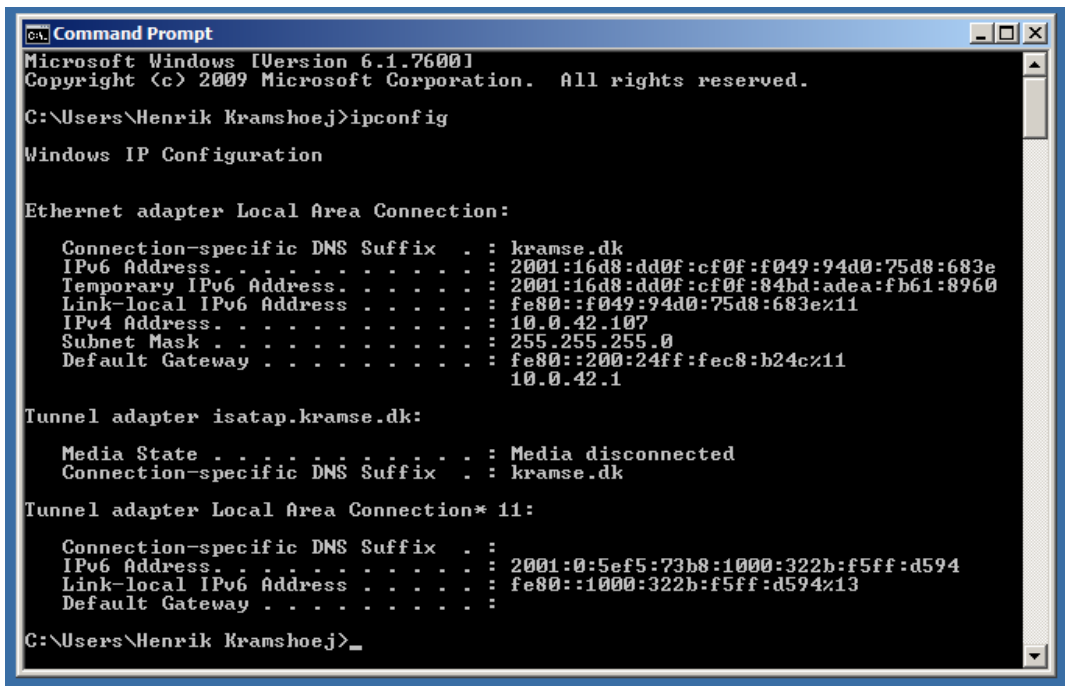
Source: https://datatracker.ietf.org/doc/html/rfc791 and updated later

IPv6 header - RFC-1883 December 1995

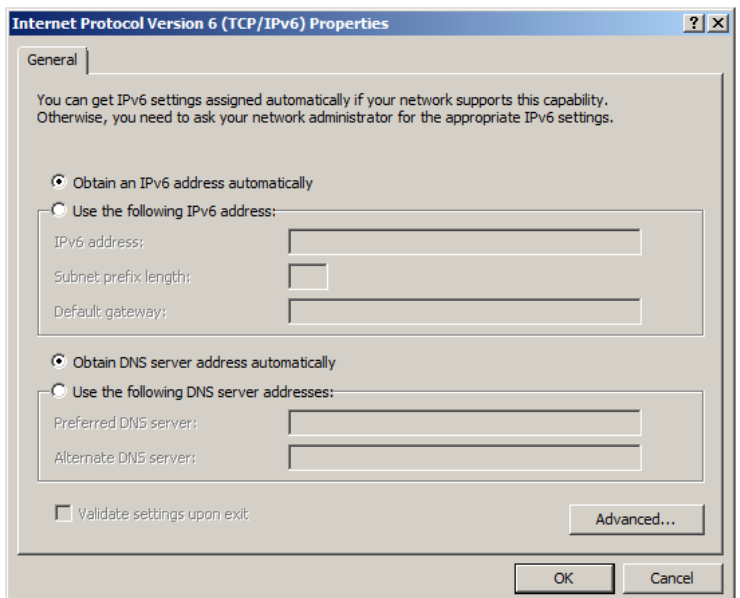




Windows - ipconfig

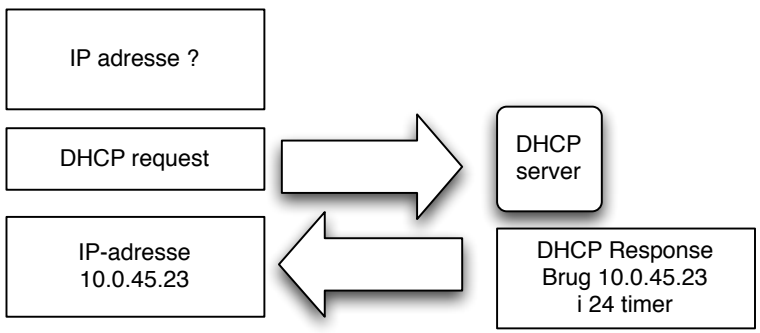


Windows – control panel with DHCP



DHCP is responsible for giving you a dynamic address

Dynamic Host Configuration Protocol (DHCP)



How does a system get the address?

Typically in IPv4 they will get them from DHCP

System sends a DHCP request, server allocates an address *lease*

IPv6 typically uses Router Advertisement – get prefix, configure their own IPv6 address

Unix - practical examples ifconfig and ping



```
$ ifconfig en0
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        inet6  fe80::216:cbff:feac:1d9f%en0 prefixlen 64 scopeid 0x4
        inet  10.0.42.15 netmask 0xffffff00 broadcast 10.0.42.255
        inet6  2001:16d8:dd0f:cf0f:216:cbff:feac:1d9f prefixlen 64 autoconf
        ether 00:16:cb:ac:1d:9f
        media: autoselect (1000baseT <full-duplex>) status: active

$ ping6 ::1
PING6(56=40+8+8 bytes) ::1 --> ::1
16 bytes from ::1, icmp_seq=0 hlim=64 time=0.089 ms
16 bytes from ::1, icmp_seq=1 hlim=64 time=0.155 ms

$ traceroute6 2001:16d8:dd0f:cf0f::1
traceroute6 to 2001:16d8:dd0f:cf0f::1 (2001:16d8:dd0f:cf0f::1)
from 2001:16d8:dd0f:cf0f:216:cbff:feac:1d9f, 64 hops max, 12 byte packets
 1  2001:16d8:dd0f:cf0f::1  0.399 ms  0.371 ms  0.294 ms
```

ICMP and Ping



Internet Control Message Protocol (ICMP)

Error conditions are signalled using this

The ping program sends ICMP ECHO request and expect ICMP ECHO reply

```
$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1): 56 data bytes
64 bytes from 192.168.1.1: icmp_seq=0 ttl=150 time=8.849 ms
64 bytes from 192.168.1.1: icmp_seq=1 ttl=150 time=0.588 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=150 time=0.553 ms
```

traceroute



traceroute works using the Time to Live (TTL) or Hop-Count fields

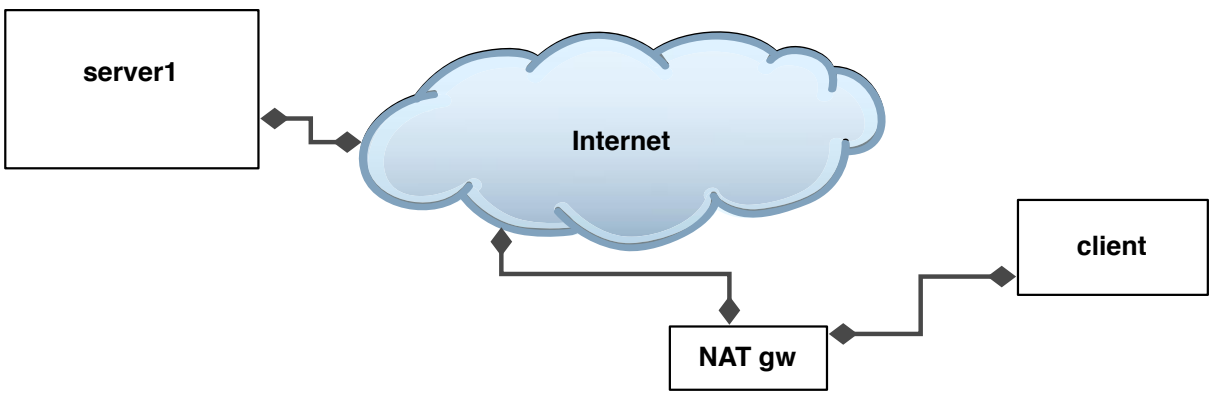
Each router will subtract one from this, and if zero – return ICMP message

Unix uses UDP with a high port number as destination, receives ICMP Port Unreachable

The tool is called `tracert` on Microsoft Windows and uses ICMP Echo request

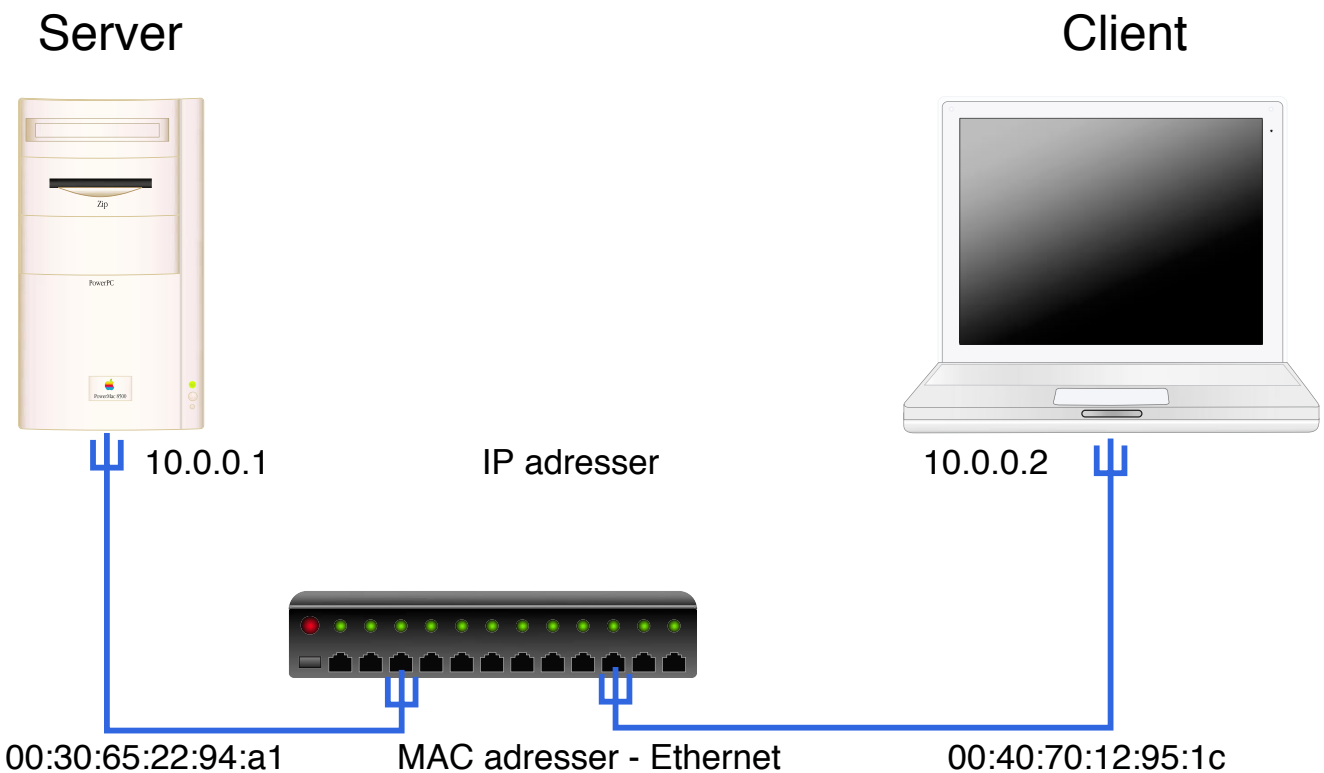
```
$ traceroute 185.129.60.129
traceroute to 185.129.60.129 (185.129.60.129),
30 hops max, 40 byte packets
 1  safri (10.0.0.11)  3.577 ms  0.565 ms  0.323 ms
 2  router (185.129.60.129)  1.481 ms  1.374 ms  1.261 ms
```

NAT Network Address Translation



- NAT is used for connecting private networks to the Internet
- NAT gateway replaces source address and forwards packets
- A quick and dirty fix that keeps messing up networks and protocols
- The NAT router/firewall has state tables

ARP in IPv4


Server
Client
10.0.0.1
IP adresser
10.0.0.2
00:30:65:22:94:a1
MAC adresser - Ethernet
00:40:70:12:95:1c

ARP request and reply



ping 10.0.0.2 from server

ARP Address Resolution Protocol request/reply:

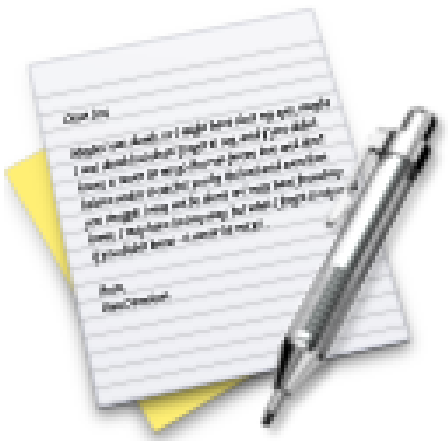
- ARP request broadcasted on layer 2 - Who has 10.0.0.2 Tell 10.0.0.1
- ARP reply (from 10.0.0.2) 10.0.0.2 is at 00:40:70:12:95:1c

IP ICMP request/reply:

- Echo (ping) request from 10.0.0.1 to 10.0.0.2
- Echo (ping) reply from 10.0.0.2 to 10.0.0.1
- ...

ARP is performed on Ethernet before IP can be transmitted

Exercise



Now lets do the exercise

Using ping and traceroute 10 min

which is number **6** in the exercise PDF.

Exercise



Now lets do the exercise

ℹ DNS and Name Lookups 10 min

which is number **7** in the exercise PDF.

Break

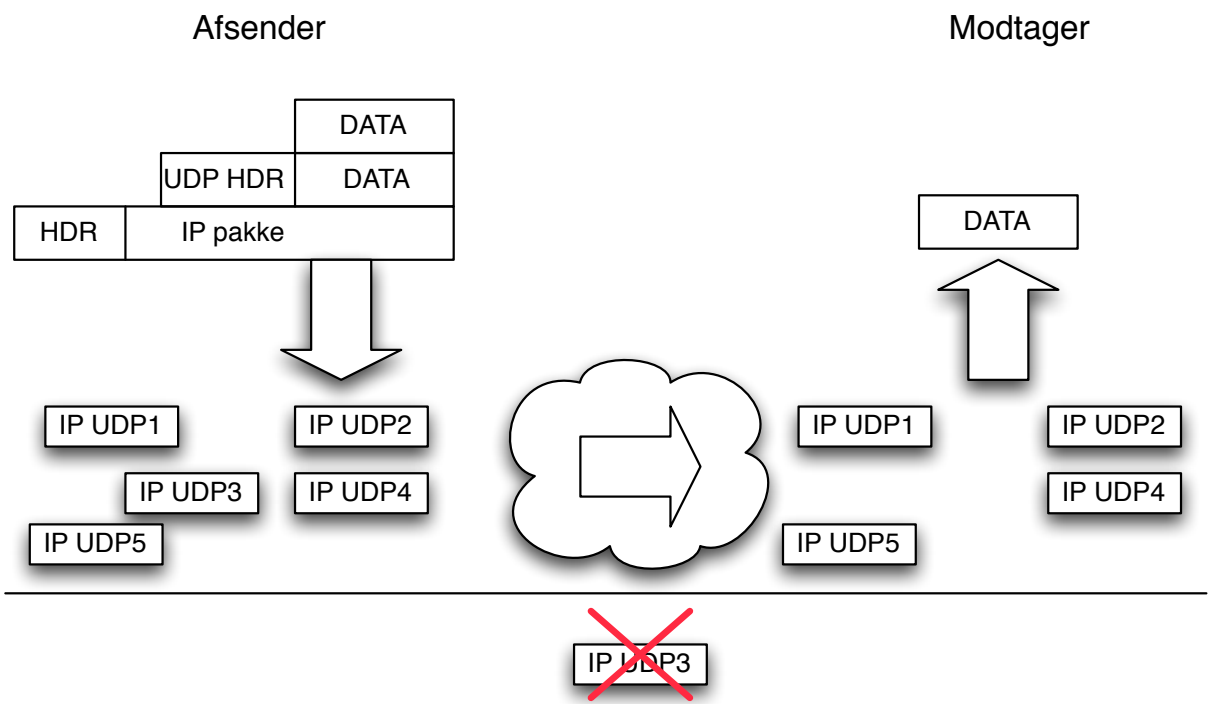


Photo by Alexas Fotos on Unsplash

Session 1: Diving deeper into networking and TCP/IP

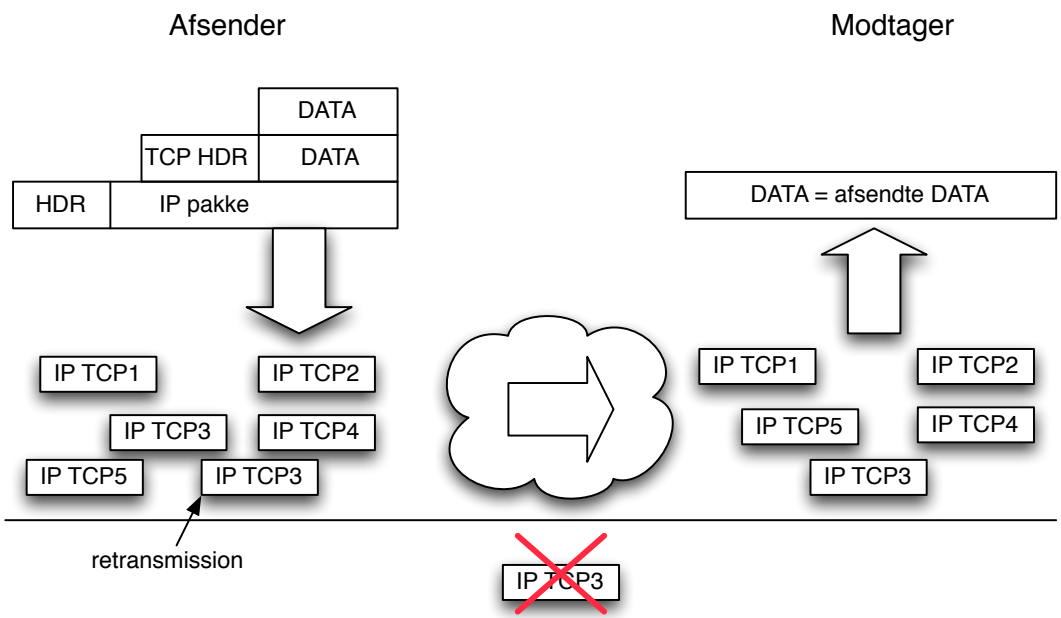


User Datagram Protocol (UDP)



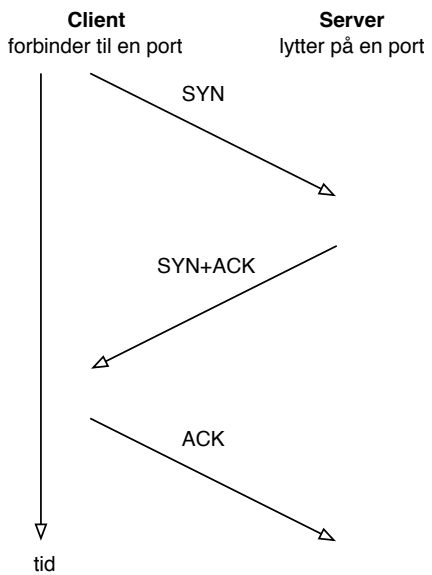
Connectionless https://en.wikipedia.org/wiki/User_Datagram_Protocol

Transmission Control Protocol (TCP)



Connection-oriented https://en.wikipedia.org/wiki/Transmission_Control_Protocol

TCP three way handshake



- Session setup is used in some protocols
- Other protocols like HTTP/2 can perform request in the first packet

Well-Known Port Numbers





IANA maintains a list of magical numbers in TCP/IP

Lists of protocl numbers, port numers etc.

A few notable examples:

- Port 25/tcp Simple Mail Transfer Protocol (SMTP)
- Port 53/udp and 53/tcp Domain Name System (DNS)
- Port 80/tcp Hyper Text Transfer Protocol (HTTP)
- Port 443/tcp HTTP over TLS/SSL (HTTPS)

Source: http://www.iana.org

Whois – Where do IP addresses come from



A these magical numbers we use on the internet are administered by IANA https://www.iana.org/

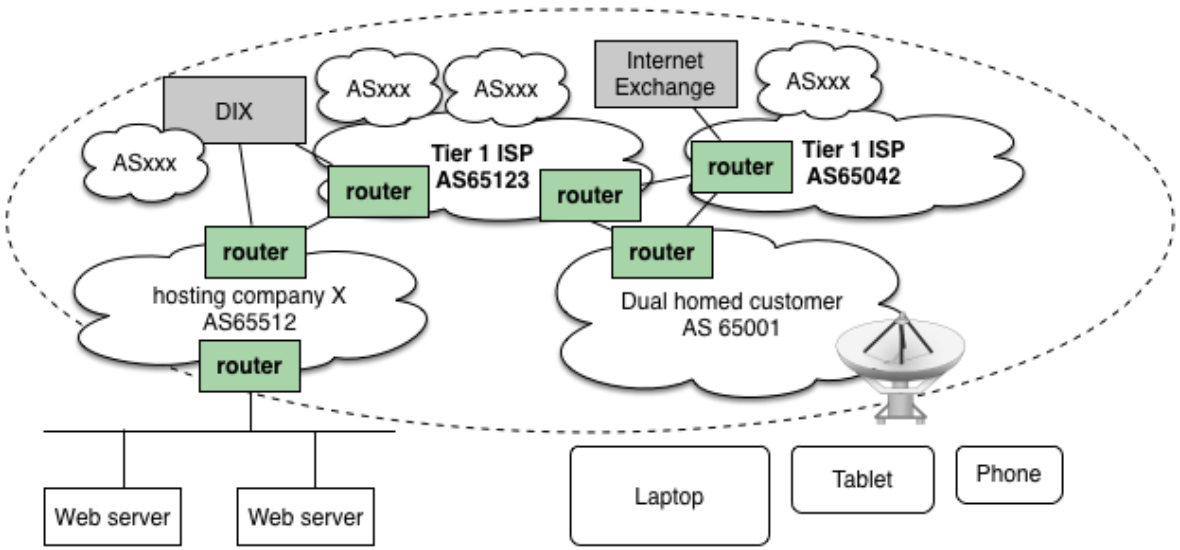
They have handed out portions to the Region Internet Registries (RIR)

- RIPE (Réseaux IP Européens) http://ripe.net
- ARIN American Registry for Internet Numbers http://www.arin.net
- Asia Pacific Network Information Center http://www.apnic.net
- LACNIC (Regional Latin-American and Caribbean IP Address Registry) - Latin America and some Caribbean Islands

AFRINIC https://afrinic.net/

They are memberbased, and members are called Local Internet Registries (LIRs) og National Internet Registry (NIR)

Hosting and internet providers



- BGP networks are used for all of the Internet
- New standards like Resource Public Key Infrastructure (RPKI) are underway
- Try RIPE BGPlay https://stat.ripe.net/special/bgplay#bgplay_fetch.resource=185.129.60.1

Routing Basics: Prefixes, AS and BGP



- **Prefix** – a block of addresses, like 192.0.2.0/24 or 2001:db8:1234::/48
- **Autonomous System (AS)** – a network under one routing policy, identified by an AS number (ASN) registered at an RIR such as RIPE NCC
- **BGP** – Border Gateway Protocol, how autonomous systems tell each other which prefixes they can reach
- **RPKI and ROA** – signed statements of which AS may originate a prefix, prefixes without a valid ROA are filtered by many networks
- **Upstream / transit** – the provider that announces your prefixes to the rest of the internet
- **Smallest routable prefix** – normally /24 for IPv4 and /48 for IPv6
- Using your own PI addresses requires an ASN, a BGP session and an upstream willing to announce them – usually a business connection.
- Exception: Netnørden offers BGP on private connections https://netnoerden.dk/faq

Public IP address: Provider Aggregatable (PA) Addresses



> PA addresses are assigned from an LIR's allocation and are registered in the RIPE Database by the LIR. The advantage of PA addresses is that the routing information for many customers can be aggregated once it leaves the provider's routing domain.

Source: RIPE Network Coordination Centre (RIPE NCC) – FAQ: ISPs
https://www.ripe.net/membership/member-support/faqs/isp-related-questions/pa-pi/

- Assigned from a Local Internet Registry's (LIR) own allocation
- Registered in the RIPE Database by the LIR itself
- Routing information for many customers can be **aggregated** into a single announcement, improving global routing efficiency
- If an end user changes their upstream provider, the PA addresses **must be returned** – they cannot be taken along
- **Usually one IPv4 for your router, or shared NAT (CGNAT) – you are here**

Public IP address: Provider Independent (PI) Addresses



> PI address space is assigned separately and not from an LIR's PA allocation. All PI assignments are registered in the RIPE Database by the RIPE NCC at the time they are assigned. PI assignments are usually small; they cannot be aggregated into larger blocks. The disadvantage of this is that network operators throughout the Internet may choose not to route them.

Source: RIPE Network Coordination Centre (RIPE NCC) – FAQ: ISPs
https://www.ripe.net/membership/member-support/faqs/isp-related-questions/pa-pi/

- Assigned **directly by the RIPE NCC** (a Regional Internet Registry), not from an LIR's allocation
- Registered in the RIPE Database by the RIPE NCC at assignment time
- Usually small blocks that **cannot be aggregated** into larger ones
- The end user **retains** the address block even when changing ISPs
- Network operators across the Internet may choose **not to route** PI assignments due to lack of aggregation

Current Status IP addresses IPv4 vs IPv6

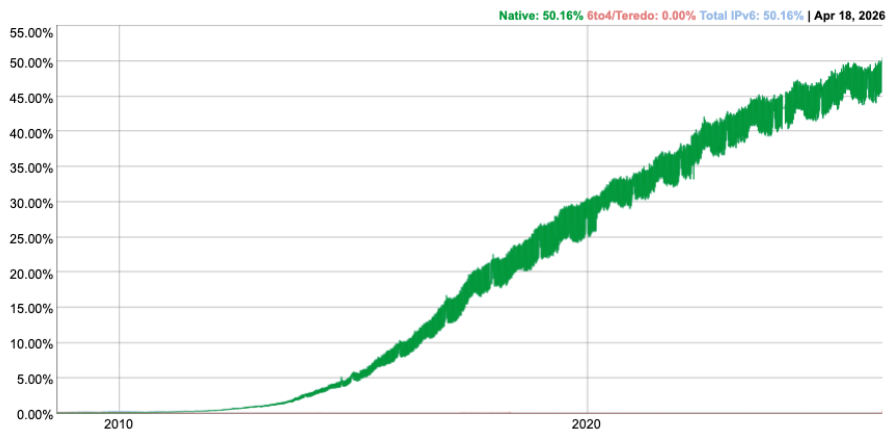


As the RIPE NCC has run out of IPv4 addresses, no new IPv4 PI address space can be assigned. IPv6 PI address space is still available.

- **IPv4 PI:** No longer available – RIPE NCC has exhausted its IPv4 pool, can be bought around 20-40USD/IP in chunks of at least 256
- **IPv4 PA:** Limited availability – RIPE NCC has exhausted its IPv4 pool, providers are charging per IP
  Usually in Denmark you would get a single public IP for your home connection and enterprise customers can pay for more
- **IPv6 PI: Still available for assignment through a sponsoring LIR**
- **IPv6 PA:** Available through LIR – but cannot be migrated to new provider!

So yes, I consider IPv6 a pre-requisite for digital sovereignty!

IPv6 in Denmark and the EU



"Now, 18 years since Google started recording this data, access via native IPv6 has for the first time exceeded 50% (50.10% on 28 March 2026, to be precise)."

Source: https://pulse.internetsociety.org/en/blog/2026/04/18-years-later-ipv6-reaches-majority/

The cost of public IP addresses



RIPE NCC members pay an annual contribution (service fee) per Local Internet Registry (LIR). For 2026, this fee is €1,800.
This is charged on a pro rata basis for new members. New members and existing members registering additional LIR accounts also pay a sign-up fee, which is currently €1,000.
Members can have more than one LIR account, but they must pay the membership and sign-up fee for each LIR account. Members also pay €75 per independent assignment or legacy resource and €50 per ASN assignment.

Source: https://www.ripe.net/membership/payment/

- Your own PA allocation requires becoming a RIPE NCC member (LIR) – expensive
- IPv6 PI is cheap, even a private person can request a /48 through a sponsoring LIR for about EUR 100 per year
- Using PI still requires an ASN and an upstream that runs BGP with you

Full-disclosure: I have a company that does these requests and charge a yearly fee of €100 – where €75 is paid to RIPE

IPv6 Neighbor Discovery Protocol (NDP)



OSI	IPv4	IPv6
Network	IP / ICMP	IPv6 / ICMPv6
Link	ARP	
Physical	Physical	Physical

Address Resolution Protocol (ARP) is replaced

NDP expands on the ARP concept, similar command `arp -an` compared to `ndp -an`

Can do some things we knew from DHCPv4 still DHCPv6 exist

Note ICMPv6 often need to be added to firewall rules for NDP! 🚩

ARP vs NDP



So at the low level, near the hardware we have protocols connecting IP addreses with MAC addresses, Ethernet and Wi-Fi are commonly found

```
hlk@bigfoot:basic-ipv6-new$ arp -an
? (10.0.42.1) at 0:0:24:c8:b2:4c on en1 [ethernet]
? (10.0.42.2) at 0:c0:b7:6c:19:b on en1 [ethernet]

hlk@bigfoot:basic-ipv6-new$ ndp -an
Neighbor                        Linklayer Address  Netif Expire    St Flgs Prbs
::1                             (incomplete)         lo0 permanent R
2001:16d8:ffd2:cf0f:21c:b3ff:fec4:e1b6 0:1c:b3:c4:e1:b6 en1 permanent R
fe80::1%lo0                     (incomplete)         lo0 permanent R
fe80::200:24ff:fec8:b24c%en1  0:0:24:c8:b2:4c      en1 8h54m51s  S  R
fe80::21c:b3ff:fec4:e1b6%en1  0:1c:b3:c4:e1:b6     en1 permanent R
```

ARP and NDP problems



- This mapping is used in your operating system, keep a dynamic ARP/neighbor cache – a table
- Switches map devices to ports – tables
- Routers remember your IP, so it can send responses back – tables
- A table has a maximum size! This can cause problems 🚩
- This is all done without ANY security – you can lie, attackers can lie
- See ARP spoofing and a sample tool https://en.wikipedia.org/wiki/ARP_spoofing and https://en.wikipedia.org/wiki/DSniff

IPv6 addressing RFC-4291



Addresses are always 128-bit identifiers for interfaces and sets of interfaces

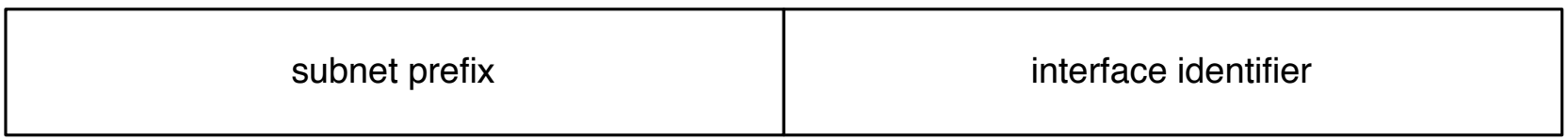
Unicast: An identifier for a **single interface**.
A packet sent to a unicast address is delivered to the interface identified by that address.

Anycast: An identifier for a **set of interfaces** (typically belonging to different nodes).
A packet sent to an anycast address is **delivered to one** of the interfaces identified by that address (the "nearest"one, according to the routing protocols' measure of distance).

Multicast: An identifier for a **set of interfaces** (typically belonging to different nodes).
A packet sent to a multicast address is **delivered to all interfaces identified by that address**.

IPv6 addressing RFC-4291, cont.





2001:16d8:ff00:012f:0000:0000:0000:0002
2001:16d8:ff00:12f::2

8 times 4 hex-digits seperated by colon x:x:x:x:x:x:x:x

Written as ipv6-address/prefix-length CIDR notation

Leading zeros can be removed

One or more groups of 16 bits of zeros can be replaced by ::

Examples:



- ABCD:EF01:2345:6789:ABCD:EF01:2345:6789
- Adddress 2001:DB8:0:0:8:800:200C:417A
- Address of loopback ::1
- IPv6 prefix 2a02:09d0:95::1/64, subnet 2a02:09d0:0095:0000::/64
- Address 2a02:09d0:95::1 or 2a02:09d0:0095:0000:0000:0000:0000:0001

IPv6 address - special prefixes



- link-local unicast addresses
  fe80::/10 generated from the interface MAC address EUI-64
- FEC0::/10 site-local - deprecated in RFC-3879
- 2001:0DB8::/32 NON-ROUTABLE range to be used for documentation purpose RFC-3849.
- FC00::/7 Unique Local IPv6 Unicast Addresses RFC-4193
  http://www.simpledns.com/private-ipv6.aspx
  If you do not like to put public addresses on internal network - use this instead

Hello neighbors



```
$ ping6 -w -I en1 ff02::1
PING6(72=40+8+24 bytes) fe80::223:6cff:fe9a:f52c%en1 --> ff02::1
30 bytes from fe80::223:6cff:fe9a:f52c%en1: bigfoot
36 bytes from fe80::216:cbff:feac:1d9f%en1: mike.kramse.dk.
38 bytes from fe80::200:aaff:feab:9f06%en1: xrx0000aaab9f06
34 bytes from fe80::20d:93ff:fe4d:55fe%en1: harry.local
36 bytes from fe80::200:24ff:fec8:b24c%en1: kris.kramse.dk.
31 bytes from fe80::21b:63ff:fef5:38df%en1: airport5
32 bytes from fe80::216:cbff:fec4:403a%en1: main-base
44 bytes from fe80::217:f2ff:fee4:2156%en1: Base Station Koekken
35 bytes from fe80::21e:c2ff:feac:cd17%en1: arnold.local
```

IPv6 autoconfiguration



`Modified EUI-64 format-based interface identifiers`

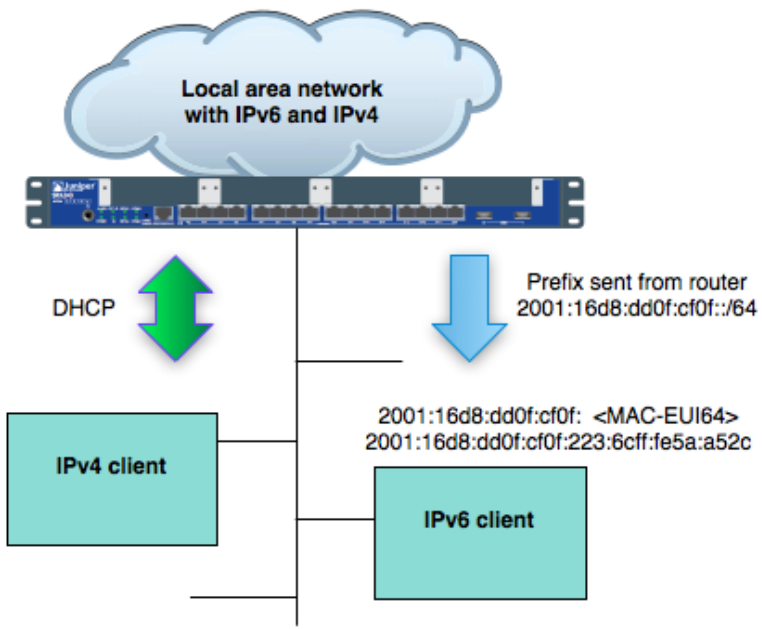
```
ifconfig en1
en1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        ether 00:23:6c:9a:f5:2c
                00-23-6c-ff-fe-9a-f5-2c  48-bit MAC stretched to become EUI-64
                02-23-6c-ff-fe-9a-f5-2c  inverting the "u" bit (universal/local bit)
                fe80::  +  0223:6cff:fe9a:f52c   add link-local prefix
        inet6 fe80::223:6cff:fe9a:f52c%en1 prefixlen 64 scopeid 0x6
```

DHCPv6 is available, but **stateless autoconfiguration** is king

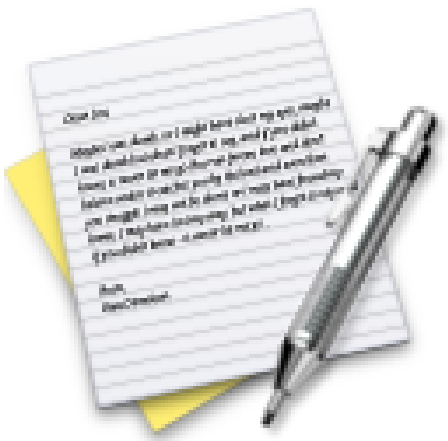
Routers announce subnet prefix via **router advertisements**

Individual nodes then combine this with their EUI64 identifier

Router Advertisement ICMPv6



Exercise



Now lets do the exercise

Whois databases 15 min

which is number **8** in the exercise PDF.

Exercise



Now lets do the exercise

IP address research up to 30 min

which is number 9 in the exercise PDF.

Break



Photo by Alexas Fotos on Unsplash

Session 2: Try out a more exercises, play with protocols, see devices



Virtual Hosts in Web Servers



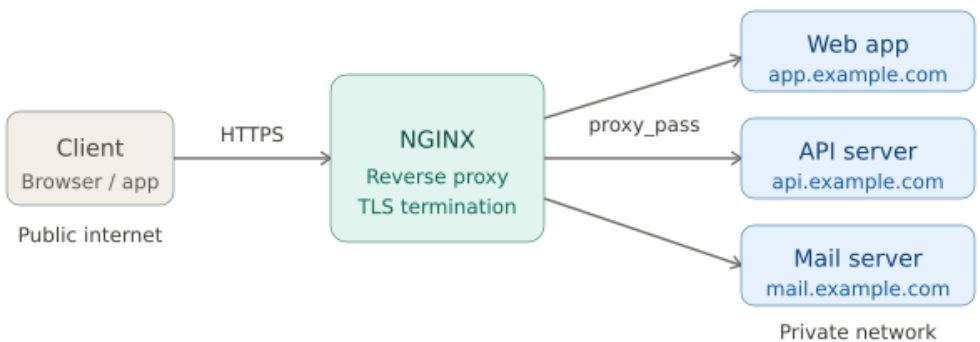
```
server { server_name example.com www.example.com; root /var/www/example.com; }
server { server_name example.org; root /var/www/example.org; }
```

- One server with one IP address can serve many websites
- DNS for both **example.com** and **example.org** points to the same IP
- The browser sends the name it wants in the HTTP **Host** header, the web server picks the matching site
- With HTTPS the name is also sent in TLS as **SNI**, so the server can present the right certificate

Side note: Digital Ocean and NGINX as a reverse proxy

That said DigitalOcean has very good documentation for lots of things! example:
https://www.digitalocean.com/community/tutorials/how-to-configure-nginx-as-a-reverse-proxy-on-ubuntu-22-04

NGINX as a Reverse Proxy



- A reverse proxy sits in front of backend servers, accepting client requests and forwarding them internally
- NGINX uses the `proxy_pass` directive to route traffic to upstream services by hostname or IP
- Can provide TLS termination, load balancing, caching, and request buffering at the edge
  A single system with NGINX get certificates with Let's Encrypt (or alternatives)

The basic tools for looking at packets

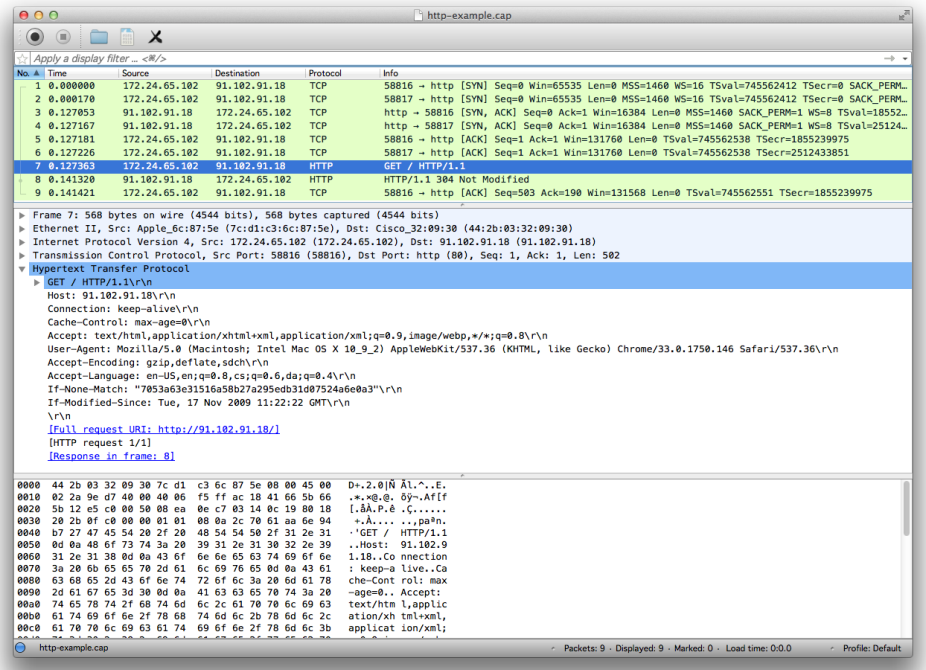


Knowledge and insight

- Networks have end-points and conversations on multiple layers
- Name resolution includes low level MAC addresses, and IP – names
- Tcpdump format, built-in to many network devices
- Wireshark is advanced, try right-clicking different places
- Remote packet dumps, like `tcpdump -i eth0 -w packets.pcap`
- Story: tcpdump was originally written in 1988 by Van Jacobson, Sally Floyd, Vern Paxson and Steven McCanne who were, at the time, working in the Lawrence Berkeley Laboratory Network Research Group https://en.wikipedia.org/wiki/Tcpdump

You can learn a lot by just listening!

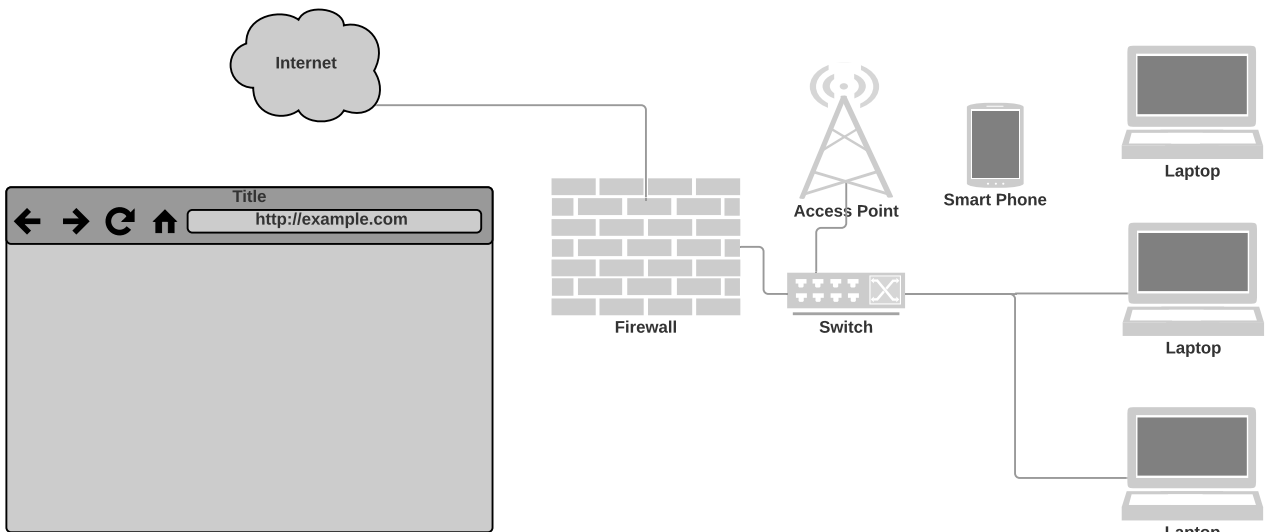
Wireshark - graphical network sniffer



Capture - Options, select a network interface

http://www.wireshark.org

Your Privacy



- Your data travels far
- Often crossing borders, virtually and literally

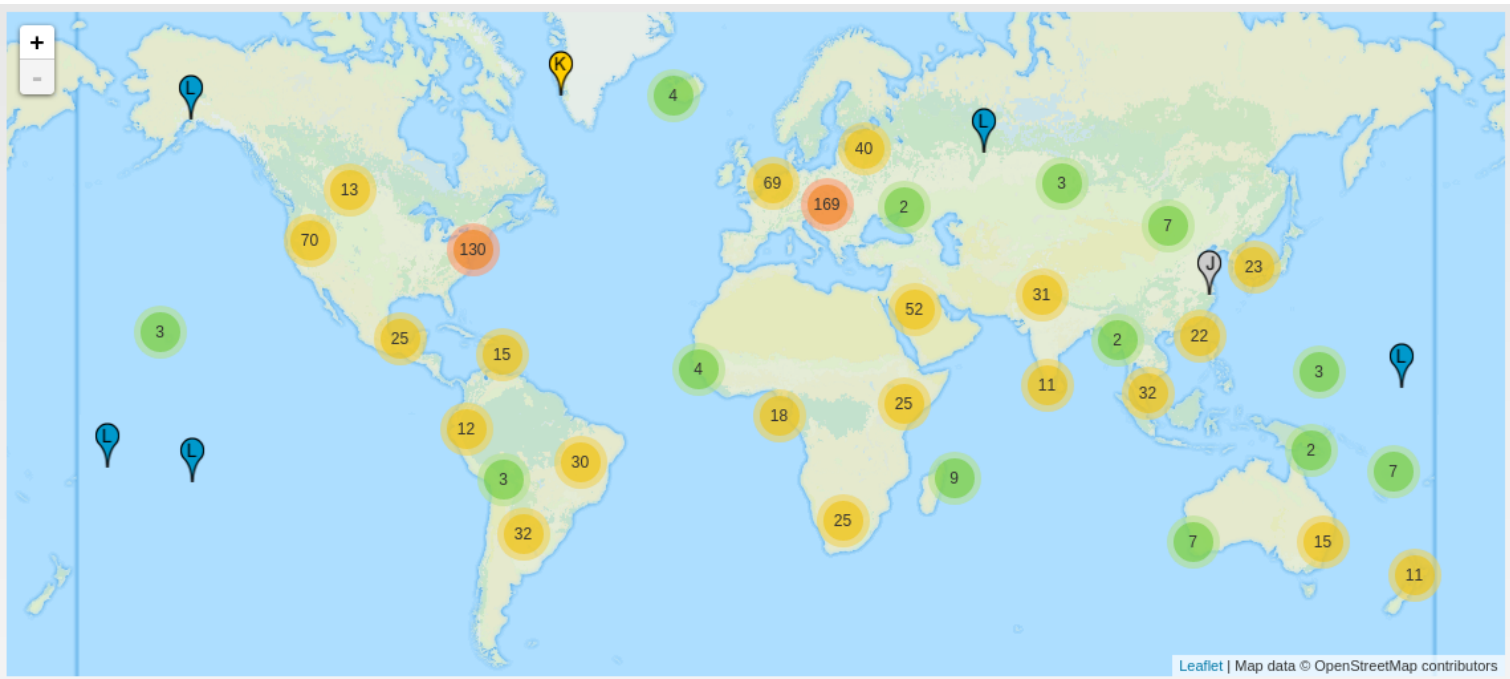
DNS root – the server addresses



```
a.root-servers.net 198.41.0.4, 2001:503:ba3e::2:30 Verisign, Inc.
b.root-servers.net 170.247.170.2, 2801:1b8:10::b University of Southern California,
Information Sciences Institute
c.root-servers.net 192.33.4.12, 2001:500:2::c Cogent Communications
d.root-servers.net 199.7.91.13, 2001:500:2d::d University of Maryland
e.root-servers.net 192.203.230.10, 2001:500:a8::e NASA (Ames Research Center)
f.root-servers.net 192.5.5.241, 2001:500:2f::f Internet Systems Consortium, Inc.
g.root-servers.net 192.112.36.4, 2001:500:12::d0d US Department of Defense (NIC)
h.root-servers.net 198.97.190.53, 2001:500:1::53 US Army (Research Lab)
i.root-servers.net 192.36.148.17, 2001:7fe::53 Netnod
j.root-servers.net 192.58.128.30, 2001:503:c27::2:30 Verisign, Inc.
k.root-servers.net 193.0.14.129, 2001:7fd::1 RIPE NCC
l.root-servers.net 199.7.83.42, 2001:500:9f::42 ICANN
m.root-servers.net 202.12.27.33, 2001:dc3::35 WIDE Project
```

Source: https://www.iana.org/domains/root/servers, see also https://en.wikipedia.org/wiki/Root_name_server

DNS root servers



As of 2019-01-29, the root server system consists of 933 instances operated by the 12 independent root server operators.

http://root-servers.org/

Run your own DNS – Unbound



Unbound is a validating, recursive, caching DNS resolver. It is designed to be fast and lean and incorporates modern features based on open standards.

To help increase online privacy, Unbound supports DNS-over-TLS which allows clients to encrypt their communication. In addition, it supports various modern standards that limit the amount of data exchanged with authoritative servers.

https://www.nlnetlabs.nl/projects/unbound/about/

My preferred local recursive DNS server

Linux Wireguard VPN

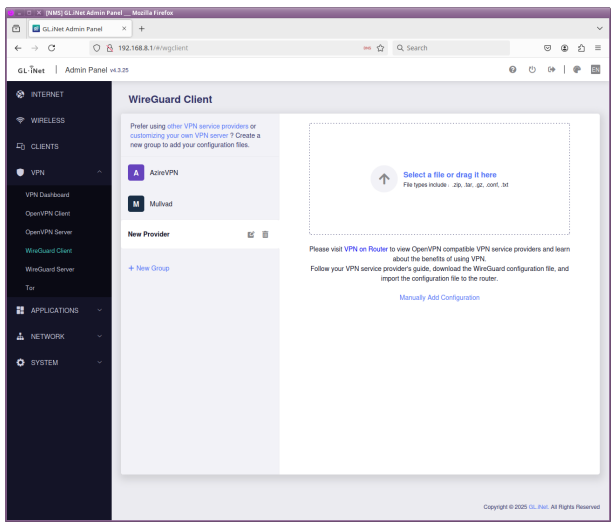


> WireGuard is a secure network tunnel, operating at layer 3, implemented as a kernel virtual network interface for Linux, which aims to replace both IPsec for most use cases, as well as popular user space and/or TLS-based solutions like OpenVPN, while being more secure, more performant, and easier to use.

Description from https://www.wireguard.com/papers/wireguard.pdf

- Very easy to setup!
- single round trip key exchange, based on NoiseIK
- Short pre-shared static keys—Curve25519
- strong perfect forward secrecy
- Transport speed is accomplished using ChaCha20Poly1305 authenticated-encryption
- encapsulation of packets in UDP
- WireGuard can be simply implemented for Linux in less than 4,000 lines of code, making it easily audited and verified

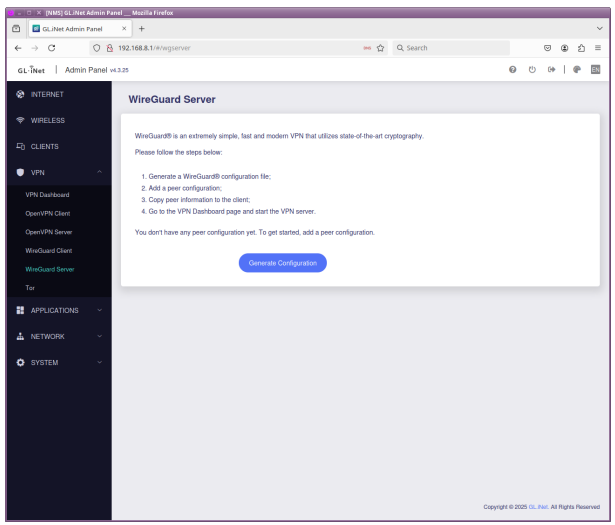
Virtual Private Network (VPN) – client



You can enable VPN functionality as a client or server. Client would be connecting you to your *home network* when travelling. Picture shown is the GL-Inet router Opal web interface

We recommend Wireguard as a modern Virtual Private Network technology

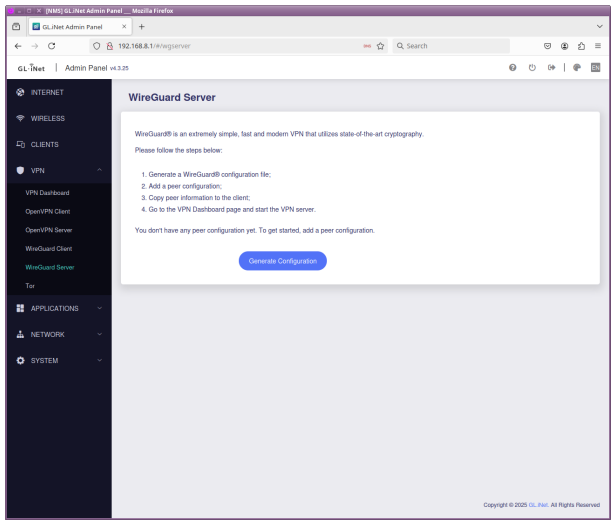
Virtual Private Network (VPN) – server



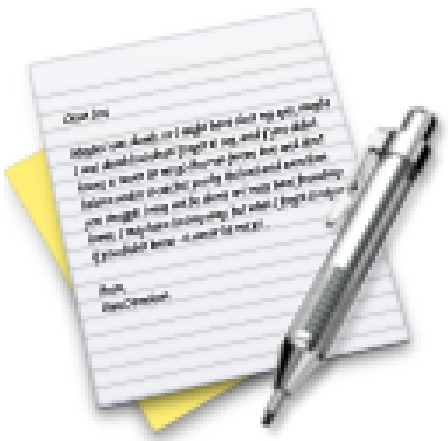
Server would be if this was your home router. Picture shown is the GL-Inet router Opal web interface

We recommend Wireguard as a modern Virtual Private Network technology

Demo: Web interface on router



Exercise

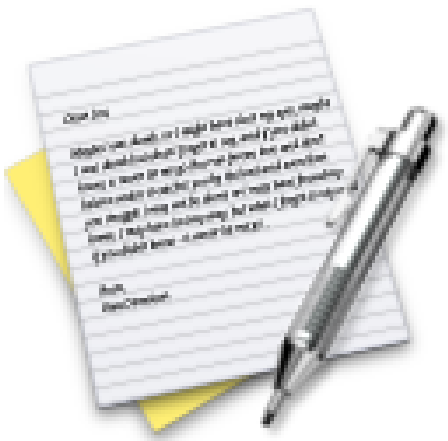


Now lets do the exercise

ℹ Wireshark and Tcpdump 15 min

which is number **10** in the exercise PDF.

Exercise



Now lets do the exercise

ℹ Nping check ports 10 min

which is number **11** in the exercise PDF.

Break

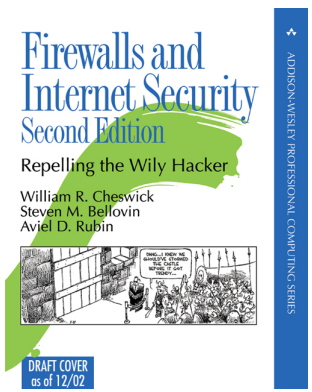


Photo by Alexas Fotos on Unsplash

Bonus: Activating Firewalls



Back in the day: Firewalls and Internet Security, 1994

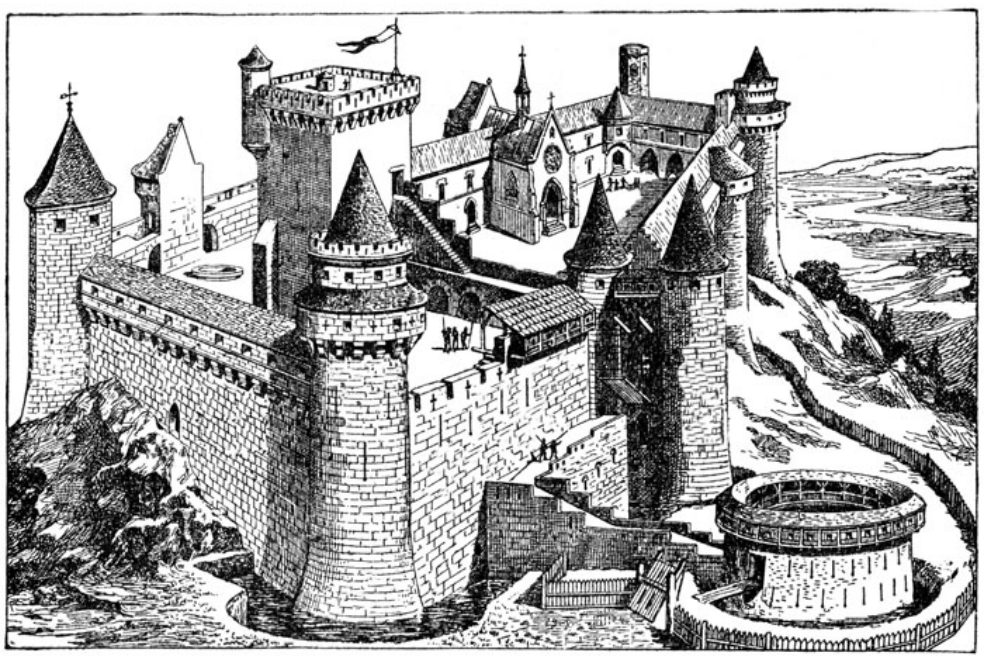


- *Firewalls and Internet Security: Repelling the Wily Hacker* , Second Edition 2003, William R. Cheswick, Steven M. Bellovin, and Aviel D. Rubin, **2003** http://www.wilyhacker.com/
- The full PDF—and the full LaTeX source of the book. Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.
- How to configure firewalls often boil down to, should we allow protocol X
- If we allow certain protocols through a firewall, we are asking for trouble

Administration and securing your own environment

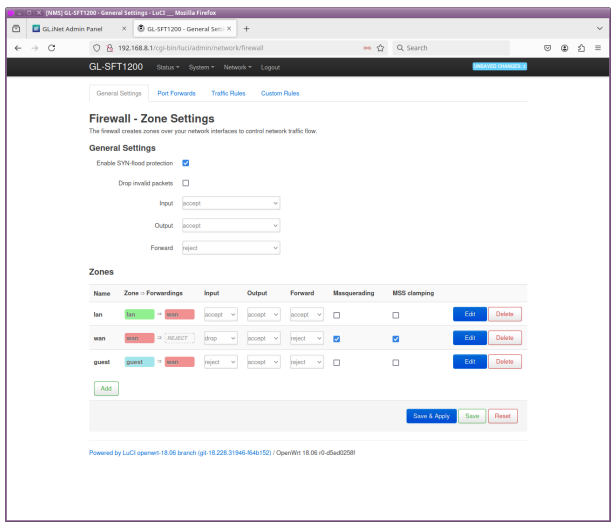


Defense in depth



Picture originally from: http://karenswhimsy.com/public-domain-images

Firewalls



- Firewalls are a great project to look into, there are so many ☺
- I recommend looking into open source firewalls like OPNsense, UFW and OpenBSD and many routers also include firewalls. You can often find images from commercial vendors, that you can try out.
- Picture is from GL-Inet router Opal with OpenWRT/LuCI which is router software for smaller routers

What is a packet filter

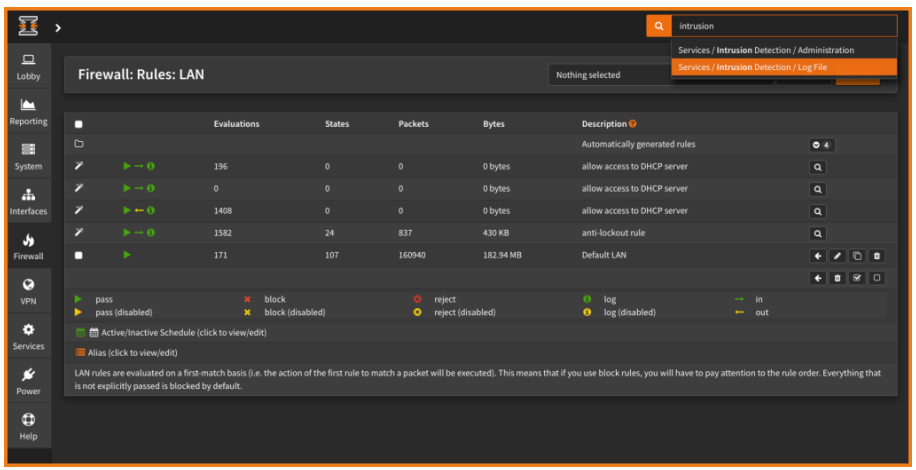


We may want to consider the difference between different types of firewalls/devices:

- Network layer, we often call them packet filters, stateless
- Application level, we often call them, stateful filtering and gateways
- Firewalls are by design a choke point, natural place
  to do network security monitoring!

They are all firewalls – or firewall devices!

Firewall OPNsense GUI based and easy to install



- OPNsense https://opnsense.org/
- Firewall built on FreeBSD with web interface
- Originally thoughts from m0n0wall and later https://www.pfsense.org/

- Danish companies have been using these for many years now

Linux Uncomplicated Firewall (UFW)



Debian / Ubuntu:

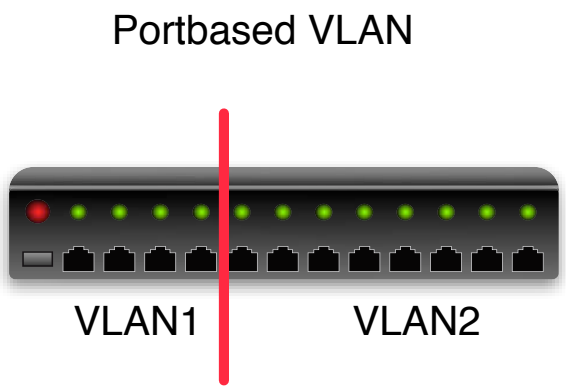
```
root@debian01:~# apt install ufw
...
root@debian01:~# ufw allow 22/tcp
Rules updated
Rules updated (v6)
root@debian01:~# ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@debian01:~# ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     ALLOW IN    Anywhere
[ 2] 22/tcp (v6)                ALLOW IN    Anywhere (v6)
```

- Extremely easy to use – I recommend and use the (Uncomplicated Firewall) UFW

Together with Firewalls - Virtual LAN (VLAN)





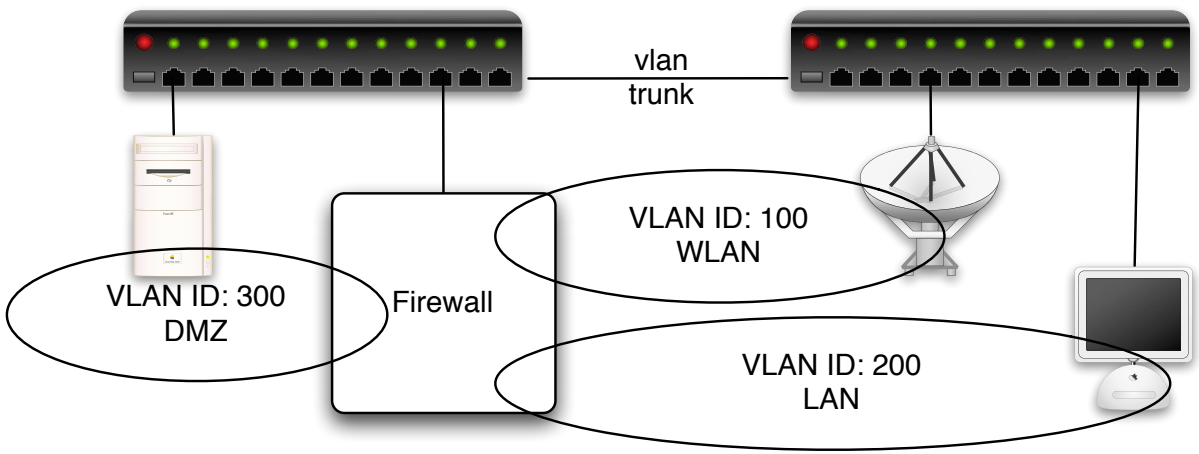
Managed switches often allow splitting into zones called virtual LANs

Most simple version is port based

Like putting ports 1-4 into one LAN and remaining in another LAN

Packets must traverse a router or firewall to cross between VLANs

Virtual LAN (VLAN) IEEE 802.1q



Using IEEE 802.1q VLAN tagging on Ethernet frames

Virtual LAN, to pass from one to another, must use a router/firewall

Allows separation/segmentation and protects traffic from many security issues

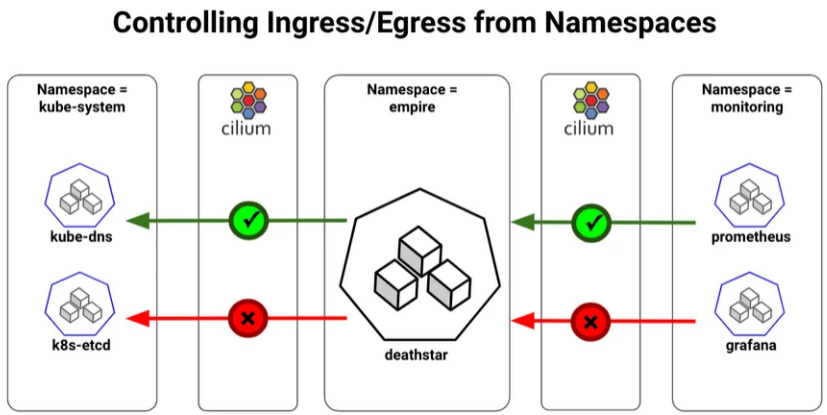
Network Segmentation: Which VLANs to create



Using VLANs to segment networks we can extend it to the rest of the network, and an example of isolated segments could be:

- Guest network for cabled and Wi-Fi clients
- Client networks can be isolated per department, floor or building
- Server networks for multiple purposes – development, staging, testing, production, manufacturing, ...
- Internet servers in a demilitarized zone (DMZ) which would make it less likely that compromised internet server would affect the rest of the network
- Dedicated printer network
- **Management network for virtualisation, one for network management, one for storage**

Cilium overview an example firewall for Kubernetes



Kubernetes provides Network Policies for controlling traffic going in and out of the pods. Cilium implements the Kubernetes Network Policies for L3/L4 level and extends with L7 policies for granular API-level security for common protocols such as HTTP, Kafka, gRPC, etc

Source: picture and text from https://cilium.io/blog/2018/09/19/kubernetes-network-policies/

Conclusion – Networks are exciting!



- Many protocols work together to provide *cat and dog pictures*
- Lots of acronyms, but a relatively small set of protocols will get you started
- Working with a small portable router and your laptop might teach you the basics

Questions?



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Email: hlk@zencurity.com

WPA3 Security



WPA3

In January 2018, the Wi-Fi Alliance announced WPA3 as a replacement to WPA2.[12][13] Certification began in June 2018,[14] and WPA3 support has been mandatory for devices which bear the "Wi-Fi CERTIFIED™"logo since July 2020.[11]

The new standard uses an equivalent 192-bit cryptographic strength in WPA3-Enterprise mode[15] (AES-256 in GCM mode with SHA-384 as HMAC), and still mandates the use of CCMP-128 (AES-128 in CCM mode) as the minimum encryption algorithm in WPA3-Personal mode.

The WPA3 standard also replaces the pre-shared key (PSK) exchange with Simultaneous Authentication of Equals (SAE) exchange, a method originally introduced with IEEE 802.11s, resulting in a more secure initial key exchange in personal mode[16][17] and forward secrecy.[18] The Wi-Fi Alliance also says that WPA3 will mitigate security issues posed by weak passwords and simplify the process of setting up devices with no display interface.[2][19]

Protection of management frames as specified in the IEEE 802.11w amendment is also enforced by the WPA3 specifications.

Source: https://en.wikipedia.org/wiki/Wi-Fi_Protected_Access

- Does not seem to be used much, in Denmark, yet – but coming

IS WPA3 supported?



> A wireless network adapter that supports Wi-Fi 6. To see if your PC supports it, check the documentation that came with it or check the PC manufacturer's website. Tip: You can also check to see if your router supports Wi-Fi 6 by opening the Command Prompt, and then typing the command netsh wlan show drivers. Look next to Radio types supported and see if it includes 802.11ax.

Source: https://support.microsoft.com/en-us/windows/faster-and-more-secure-wi-fi-in-windows-26177a28-38ed-1a8e-7e

- Your devices must support both WPA3 in both operating system and Wi-Fi drivers!
- Windows 10 and 11 does, so try upgrading drivers
- Android does since Android 10 – pretty new still
- Apple devices have support in recent versions

WPA3 on Apple Devices



- iPhone 7 or later.
- iPad 5th generation or later.
- Apple TV 4K or later.
- Apple Watch series 3 or later.
- Mac computers (late 2013 or later, with 802.11ac or later)

Source: https://support.apple.com/da-dk/guide/security/sec8a67fa93d/web

Wi-Fi CERTIFIED WPA3™



WPA3™ provides cutting-edge security protocols to the market. Building on the widespread success and adoption of Wi-Fi security, WPA3 adds new features to simplify Wi-Fi security, enable more robust authentication, deliver increased cryptographic strength for highly sensitive data markets, and maintain resiliency of mission critical networks. All WPA3 networks:

- Use the latest security methods
- Disallow outdated legacy protocols
- Require use of Protected Management Frames (PMF)

Since Wi-Fi networks differ in usage purpose and security needs, WPA3 includes additional capabilities specifically for personal and enterprise networks. Users of WPA3-Personal receive increased protections from password guessing attempts, while WPA3-Enterprise users can now take advantage of higher-grade security protocols for sensitive data networks.

WPA3 is a mandatory certification for Wi-Fi CERTIFIED™ devices.

Source: https://www.wi-fi.org/discover-wi-fi/security

WPA3-Personal



WPA3-Personal brings better protections to individual users by providing more robust password-based authentication, even when users choose passwords that fall short of typical complexity recommendations. This capability is enabled through Simultaneous Authentication of Equals (SAE). The technology is resistant to offline dictionary attacks where an adversary attempts to determine a network password by trying possible passwords without further network interaction.

- **Natural password selection**: Allows users to choose passwords that are easier to remember
- **Ease of use**: Delivers enhanced protections with no change to the way users connect to a network
- **Forward secrecy**: Protects data traffic even if a password is compromised after the data was transmitted

Source: https://www.wi-fi.org/discover-wi-fi/security

WPA3-Enterprise



WPA3-Enterprise builds upon the foundation of WPA2-Enterprise with the additional requirement of using Protected Management Frames on all WPA3 connections.

- **Authentication**: multiple Extensible Authentication Protocol (EAP) methods
- **Authenticated encryption**: minimum 128-bit Advanced Encryption Standard Counter Mode with Cipher Block Chaining Message Authentication (AES-CCMP 128)
- **Key derivation and confirmation**: minimum 256-bit Hashed Message Authentication Mode (HMAC) with Secure Hash Algorithm (HMAC-SHA256)
- **Robust management frame protection**: minimum 128-bit Broadcast/Multicast Integrity Protocol Cipher-based Message Authentication Code (BIP-CMAC-128)

Source: https://www.wi-fi.org/discover-wi-fi/security

WPA3-Enterprise with 192-bit mode



WPA3-Enterprise also offers an optional mode using 192-bit minimum-strength security protocols and cryptographic tools to better protect sensitive data.

- **Authentication**: Extensible Authentication Protocol – Transport Layer Security (EAP-TLS) using Elliptic Curve Diffie-Hellman (ECDH) exchange and Elliptic Curve Digital Signature Algorithm (ECDSA) using a 384-bit elliptic curve
- **Authenticated encryption**: 256-bit Galois/Counter Mode Protocol (GCMP-256)
- **Key derivation and confirmation**: 384-bit Hashed Message Authentication Mode (HMAC) with Secure Hash Algorithm (HMAC-SHA384)
- **Robust management frame protection**: 256-bit Broadcast/Multicast Integrity Protocol Galois Message Authentication Code (BIP-GMAC-256)

The 192-bit security mode offered by WPA3-Enterprise ensures the right combination of cryptographic tools are used and sets a consistent baseline of security within a WPA3 network.

Source: https://www.wi-fi.org/discover-wi-fi/security